RUN / SOURCE ATLAS / 03.11.2026

What makes RUN work.

A visual guide to the app, its code, its data, its borrowed material and its F-Droid path.

Version 2.0.37 · build 137

Android 12+

Read-only analysis

Source snapshot cce930915554

RUN is the product name; Ayva and Focus by Rj remain in source paths and class names.

A guide you can navigate

Start with the UI tree. Follow a feature to its files. Use the file browser when you need to go deeper.

1,109	396	13	1,878
tracked files indexed	production Kotlin files	feature maps	static source reference edges

WHEN YOU WANT TO…	GO HERE
Understand where a feature appears	UI tree
See which component does the work	Architecture + feature chapters
Trace an action across features	Six real-action flows
Understand where data is protected	Storage and key map
Find what was borrowed and its terms	Provenance + per-asset ledger
Prepare an F-Droid submission	Current → target → files → validation
Look up any file or component name	Searchable browser + directory tree

> **Reading the arrows.** Feature diagrams explain reviewed responsibilities. The file inspector’s “references” and “used by” lists are extracted static import/identifier matches, not a measured runtime call graph. Dashed certainty cannot be inferred from an arrow.

CODE WORD	PLAIN MEANING
Screen / composable	What the user sees inside a Compose UI.
Activity	An Android window or entry point; not every feature has its own Activity.
ViewModel / session	The current screen/conversation state and operations.
Repository / manager	The owner or coordinator used to read and update shared data.
DAO / database / entity	Queries, storage schema and stored record shape.
Service / receiver / worker	Background lifetime, event/alarm delivery and deferred work.
Parser / executor	Interpret input / apply a validated action. These are different responsibilities.
Asset / resource	Packaged text, art, fonts, layouts or static data; these do not all share one license.

Contents

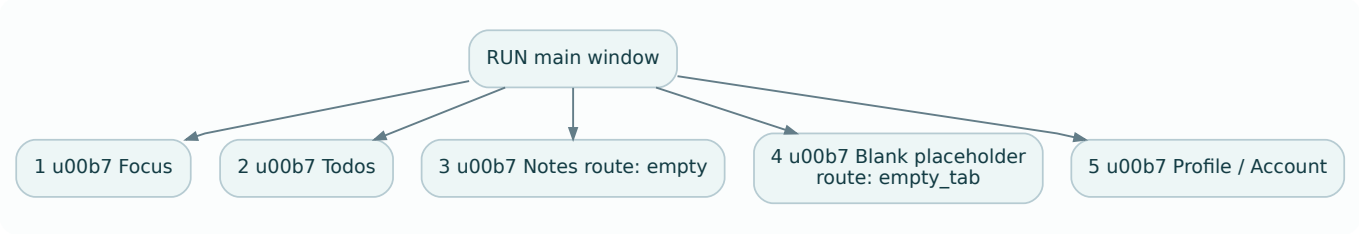
01 · Start here

02 · UI tree

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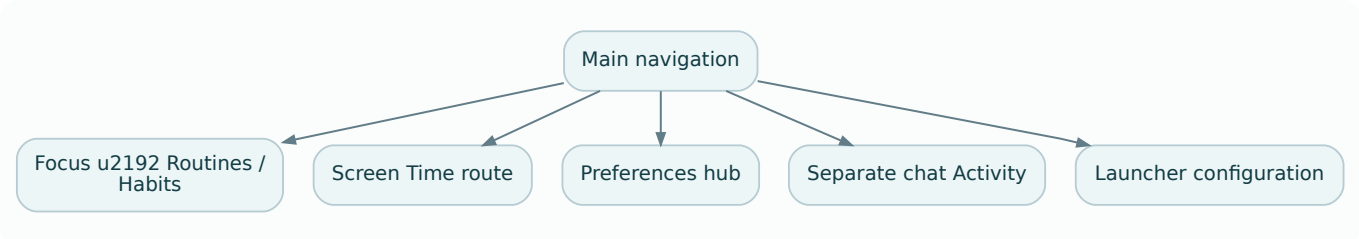
The UI tree

This is an explanation of the existing app, not a proposed redesign. Route names are included because some do not match what you see.



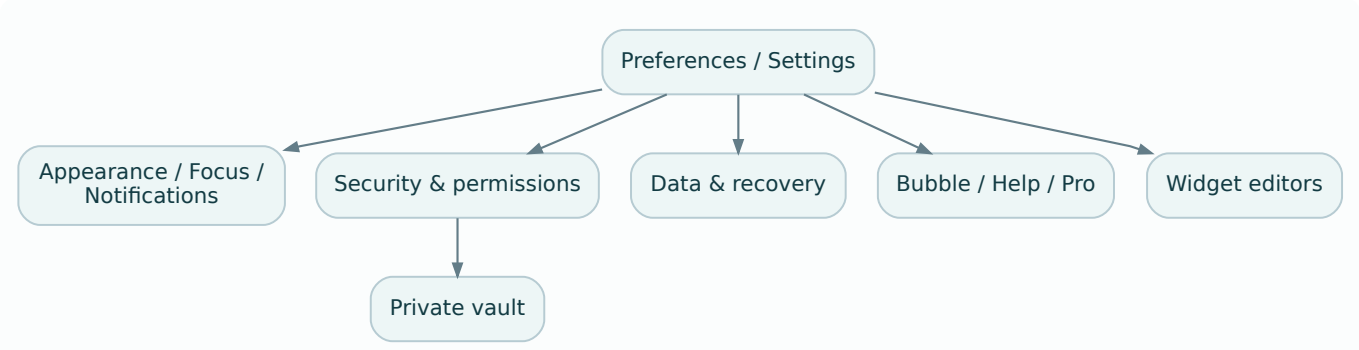
VISIBLE DESTINATION	CODE / INTERNAL ROUTE
Focus / Home	DashboardScreen.kt · dashboard
Todos	TodosScreen.kt · todos
Notes	NotesScreen.kt · empty
Blank fourth dock slot	Screen.kt · empty_tab
Profile / Account	AccountScreen.kt · account

Other surfaces



Habits and Routines are secondary pages; Screen Time has its own route. Floating chat and widget configuration are Android entry points outside the ordinary bottom-dock pages. Note sharing/quick edit, drills, voice capture, chest reveal and backup dialogs are focused flows or overlays.

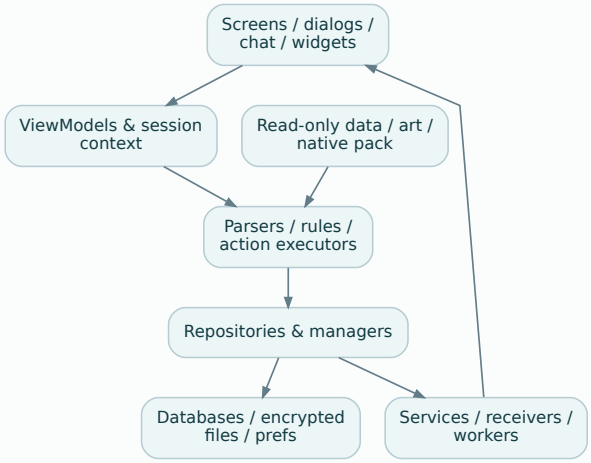
Settings branches



How the code fits together

The folder tree is about responsibilities, not an instruction to reorganize files. Several features intentionally cross folders: widgets use note/task repositories, and chat executes the same actions as the native pages.

```
Ayva/                                Repository (product: RUN)
├── app/src/main/
│   ├── java/com/focusbyrj/app/
│   │   ├── FocusApplication.kt       Startup and shared owners
│   │   ├── MainActivity.kt           Routes and main window
│   │   ├── ui/                       Screens, components, theme, speech
│   │   ├── data/                     Models, DAOs, repositories, databases
│   │   ├── util/assistant/           Context, language, planning, experiences
│   │   ├── util/backup/ + crypto/    Recovery, encryption, data boundaries
│   │   ├── service/                  Alarms, background service, overlays
│   │   └── widget/                   Launcher renderers, actions, editors
│   ├── assets/                       Dictionary, seed data, animation, rules
│   ├── res/                          Fonts, drawables, raw scenes, layouts
│   ├── jniLibs/                      Packaged native worker by CPU ABI
│   └── AndroidManifest.xml           Android components and permissions
├── app/schemas/                      Exported Room migration evidence
├── app/src/test/ + androidTest/      Host regression and device checks
├── tools/                            Importers, generators, native build, checks
├── gradle/ + *.gradle.kts             Dependencies and Android build rules
├── .github/workflows/                Validation and release automation
├── fastlane/metadata/                Store listing text and graphics
├── docs/                             Plans, audit history, design evidence
└── LICENSE + TRADEMARK.md            Source grant and separate brand policy
```



The shared-owner rule

A screen, chat message and widget can be three ways to operate on one task. Their UI state may differ, but they should not become three competing task databases. Look for the repository or canonical owner when changing behavior.

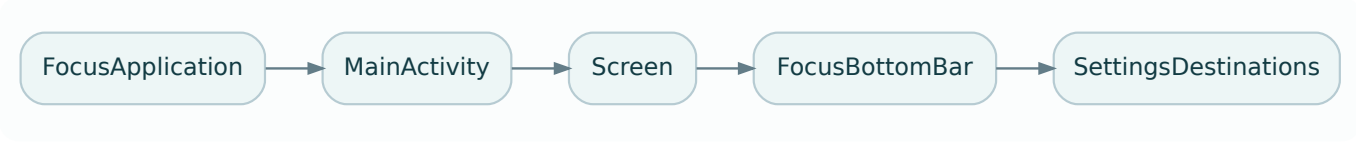
FEATURE 01 / 13

Find your way around

One activity hosts the main pages; Android entry points also open focused tools.

What you see

The bottom dock has Focus, Todos, Notes, a blank placeholder, and Profile. Habits, Routines, Screen Time and Settings are secondary destinations. Chat can also open as its own floating window.



How the parts help each other

- Startup recovery completes before ordinary data managers start.
- MainActivity supplies the repositories and ViewModels used by the visible screens.
- The top-bar profile/avatar shortcut and the bottom Profile button have different destinations: Preferences versus Account.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
FocusApplication.kt	Process startup: restore recovery first, initialize managers, configure encrypted media loading and lazy databases.
MainActivity.kt	Main Android window: wires navigation, primary screens, bottom dock and settings destinations.
Screen.kt	Route-name registry. The route named empty opens Notes; empty_tab is the blank placeholder.
FocusBottomBar.kt	Five-slot bottom dock. Its fourth slot is a blank placeholder; Habits and Routines are reached elsewhere.
SettingsDestinations.kt	Registers the shared settings routes, search wrappers and legacy-route redirects.
AyvaIcon.kt	Ayva Icon component in app shell & navigation.

> **When changing this:** The internal name empty means Notes; empty_tab is the actual blank tab. Twelve launcher aliases are alternative icons for the same app, not twelve apps or screens.

Material / infrastructure: Project navigation and manifest; UI uses AndroidX Compose and Material libraries.

FEATURE 02 / 13

Focus, app blocking & routines

Turn a focus session or recurring rule into the right blocking window.

What you see

Home focus controls, installed-app categories and restrictions, routine schedules, temporary access and Screen Time statistics. Hard/soft blocking windows appear over restricted apps.



How the parts help each other

- Restrictions and routines live in FocusDatabase; usage observations come from Android Usage Access.
- The foreground service evaluates restrictions and directly adds WindowManager overlay views.
- Timer end alarms, boot recovery and screen on/off handling keep the service independent of the visible page.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
DashboardScreen.kt	User-interface screen for Dashboard.
FocusViewModel.kt	Reactive screen state and operations for Focus.
AppRepository.kt	Data access and update boundary for App.
AppRestriction.kt	App Restriction component in focus, blocking & routines.
FocusSchedule.kt	Focus Schedule component in focus, blocking & routines.
SchedulesScreen.kt	User-interface screen for Schedules.
AddRestrictionScreen.kt	User-interface screen for Add Restriction.
FocusSessionManager.kt	Restores and tracks the focus timer deadline; timer completion does not control phone-wide DND.
FocusBlockerService.kt	Monitors foreground app usage and focus/routine state; decides when direct blocking overlays are needed.
BlockOverlayManager.kt	Builds hard/soft blocking windows using WindowManager overlays; this is the primary blocker presentation.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Keep the direct WindowManager overlay architecture. Replacing it with Activity launches would reintroduce Android background-launch restrictions. Focus no longer changes phone-wide Do Not Disturb.

Material / infrastructure: Project policy/state plus Android system usage/overlay APIs. Quotes and theme assets have separate content provenance.

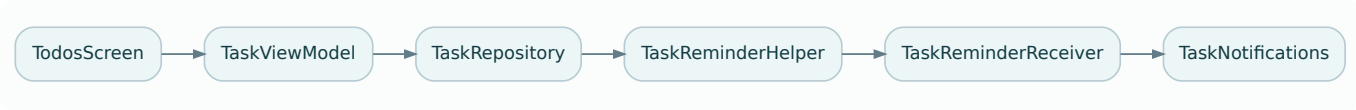
FEATURE 03 / 13

Tasks, dates & reminders

One task record powers the list, assistant commands, reminders and widgets.

What you see

Todos, quick add, recurring work, task dependencies/checklists and planned work sessions. The same task can be shown in chat or a home-screen widget.



How the parts help each other

- SmartDateParser extracts supported dates; the surrounding conversation or screen decides what the date belongs to.
- TaskRepository writes FocusDatabase and coordinates task state. Reminder helpers schedule delivery from that state.
- Reminder receivers recheck whether a task is deleted, completed or rescheduled before displaying it.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
TodosScreen.kt	User-interface screen for Todos.
QuickAddTaskActivity.kt	Android window / activity entry point for Quick Add Task.
TaskViewModel.kt	Reactive screen state and operations for Task.
Task.kt	Task component in tasks, dates & work sessions.
TaskPlanning.kt	Task Planning component in tasks, dates & work sessions.
WorkSession.kt	Work Session component in tasks, dates & work sessions.
TaskRepository.kt	Data access and update boundary for Task.
TaskReminderHelper.kt	Schedules/cancels task reminders and updates notification summaries.
TaskReminderReceiver.kt	Handles Android broadcasts or scheduled callbacks for Task Reminder.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** A parser is not a task writer. Keep task IDs, recurrence, subtasks and reminder cancellation connected when changing this area; chat, widgets and the list must agree.

Material / infrastructure: Project models and date/routing rules. Room, WorkManager and Android alarms are supporting infrastructure.

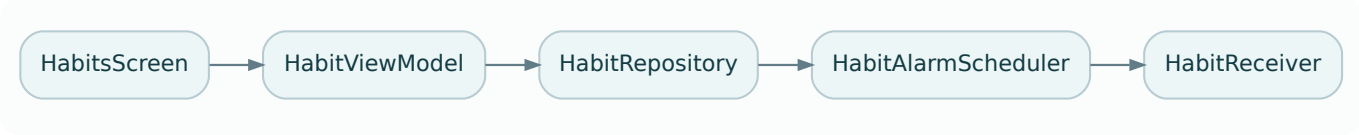
FEATURE 04 / 13

Habits & pause/resume

Track targets without treating a deliberate pause as failure.

What you see

Habit cards, targets/frequencies, logs, reminders, pause/resume controls, shields and assistant habit actions.



How the parts help each other

- Pause intervals retain the habit and real logs; streak/target calculations account for the pause.
- Pausing suppresses scheduled/queued reminders; resuming recalculates future scheduling.
- Talk habit actions pass through the assistant’s guarded editing flow and share the repository with the Habits page.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
HabitsScreen.kt	User-interface screen for Habits.
HabitsChatCard.kt	Habits Chat Card component in habits & pausing.
HabitViewModel.kt	Reactive screen state and operations for Habit.
Habit.kt	Habit component in habits & pausing.
HabitPause.kt	Pause intervals and pause-aware habit behavior; paused time is not fabricated completion.
HabitFrequency.kt	Habit Frequency component in habits & pausing.
HabitRepository.kt	Data access and update boundary for Habit.
HabitDao.kt	Database query interface for Habit.
HabitAlarmScheduler.kt	Computes and registers habit reminder windows; pause/resume affects scheduling.
HabitReceiver.kt	Handles Android broadcasts or scheduled callbacks for Habit.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** A pause is neither a completion nor deletion. Date boundaries, historical targets and reminder reconciliation must stay consistent.

Material / infrastructure: Project habit logic; habit cat artwork is in .lottie files with item-level licenses still unverified.

FEATURE 05 / 13

Notes, editor & attachments

Edit rich notes and safely connect their text, blocks and encrypted media.

What you see

Notes grid, labels, checklists, rich text, tables, sketches, image collages, audio memos, archive/private vault, sharing and PDF/DOCX exports.



How the parts help each other

- Editor widgets transform note content; the ViewModel and repository persist the model.
- NoteDatabase is SQLCipher-backed. Attachments are separate encrypted files referenced by the note.
- Coil fetches encrypted images through EncryptedMediaFetcher; share/export deliberately creates user-requested output with cleanup handling.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
NotesScreen.kt	User-interface screen for Notes.
NotesViewModel.kt	Coordinates note screen state, editing, archive/vault actions and repository operations.
KeepNoteEditor.kt	Main note-editing UI; text, blocks, formatting and attachment entry points.
KeepNoteCard.kt	Keep Note Card component in notes, editor & media.
NoteRepository.kt	Data access and update boundary for Note.
NoteEntity.kt	Stored note model, checklist/media references, privacy flags and editor content fields.
NoteDao.kt	Database query interface for Note.
NoteDatabase.kt	SQLCipher-backed note database with explicit migration/recovery handling.
NoteMediaManager.kt	Imports/manages note attachments and their storage references.
EncryptedMediaStorage.kt	Encrypts original local attachments using AES-GCM with an AndroidKeyStore-held key; legacy reads are distinct.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Changing text serialization can affect old notes, checklists, backup import and widgets. Exported documents can be plaintext by design; do not describe every file as encrypted. Notesnook-named files alone do not establish copied code.

Material / infrastructure: Project implementation; AndroidX Room, SQLCipher, Coil and bundled fonts support it. Notesnook influence needs provenance evidence before attributing a port.

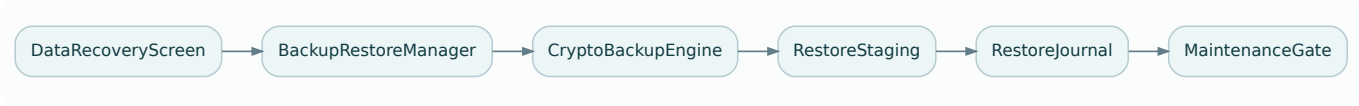
FEATURE 06 / 13

Vault, backup & recovery

Keep local protection, portable backups and crash recovery distinct.

What you see

Private notes vault, encrypted export/restore/verify, safety snapshots, saved drafts, recovery phrase and storage tools.



How the parts help each other

- Portable backups use a password-derived key; internal snapshots and local media use device-held keys. They solve different problems.
- Restore validates and stages material before replacing live data; a journal supports interrupted-operation recovery.
- FocusApplication runs pending recovery before starting ordinary writers. Participating writers use the maintenance gate.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
DataRecoveryScreen.kt	User-interface screen for Data Recovery.
BackupFlow.kt	Backup Flow component in encryption, backup & recovery.
BackupDialogs.kt	Backup Dialogs component in settings, themes & permissions.
PrivateVaultSettings.kt	Private Vault Settings component in encryption, backup & recovery.
ArchiveVaultSecurity.kt	Private archived-note PIN/recovery handling; derives wrapping keys and manages a stable vault data key.
BackupRestoreManager.kt	Collects/export/restores application data and media; validates/stages before applying live changes.
CryptoBackupEngine.kt	Portable password-protected backup envelope: current Argon2id + AES-GCM, with legacy readers.
RestoreStaging.kt	Prepares bounded validated restore content before live mutation.
RestoreJournal.kt	Crash-recovery journal: completes or rolls back interrupted restore operations at startup.
MaintenanceGate.kt	Coordinates participating writers during maintenance; not a second database or encryption algorithm.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Uninstall or a signing/package migration can destroy device-held keys. A local snapshot is not a portable backup. VaultSyncManager’s name is not evidence of an active network sync service.

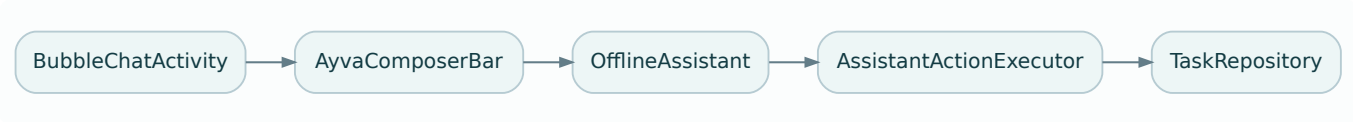
Material / infrastructure: SQLCipher Community, Argon2Kt, AndroidKeyStore and project envelope/journal code. Exact versions and notices are indexed separately.

Chat, Plan, Talk & offline experiences

Use bounded offline features and route real actions through shared app data owners.

What you see

Floating bubble/chat, Chat/Plan/Talk/Drill modes, task and habit actions, trip checklists, saved plans, dictionary, discovery, word games, riddles, jokes, facts and stories.



How the parts help each other

- The composer and mode select a supported intent; session context carries pending references and edits.
- Planning maintains draft/workspace state, then guarded executors apply changes to canonical tasks with receipts/undo. The existing planning behavior is not changed by this guide.
- The chat three-dot menu opens a bubble-icon picker using the same 12 app-icon assets. It stores the bubble choice separately and updates the existing overlay without restarting the service.
- Finite experiences and content rotation are separate from open-ended dialogue. Rotation preferences survive clearing chat. Dictionary definitions come from Wordset shards.
- Optional native ChatScript runs in a crash-isolated worker and has a Kotlin fallback. It is not an unrestricted language model.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
BubbleService.kt	Floating chat bubble: foreground lifetime, snooze/visibility, fullscreen/landscape hiding and recovery.
BubbleIconManager.kt	Shares the existing app icon artwork with the bubble; persists its separate choice and notifies the running overlay without restarting it.
BubbleIconPicker.kt	Compact chat-menu picker for the 12 existing app icons; highlights the saved bubble selection and applies on tap.
BubbleChatActivity.kt	Chat activity and integration shell: composer, message cards, actions and assistant entry point.
AyvaComposerBar.kt	Ayva Composer Bar component in chat, planning & games.
OfflineAssistant.kt	Main offline assistant turn coordinator; selects supported modes/features and guarded action proposals.
AssistantActionExecutor.kt	Guarded execution owner for assistant-requested changes and undo; response text is not write authority.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Multiple engines are not interchangeable authorities. Clearing visible history must not silently delete tasks, saved plans or rotation state. The native worker shares the app UID; a separate process is not a privilege sandbox.

Material / infrastructure: Confirmed: Wordset data and pinned ChatScript runtime/rules packaging. Project-authored finite content is documented; research candidates are not all shipped. Rive/Lottie cat artwork has separate unresolved item provenance.

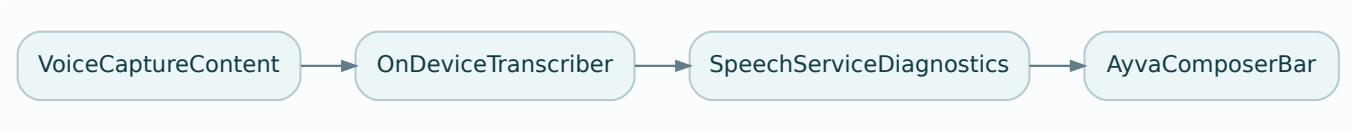
FEATURE 08 / 13

Voice capture & transcript editing

Turn a recognizer session into editable text before the assistant acts.

What you see

Speech popup, live visualization, styles and editable transcription. Notes audio recording is a separate feature.



How the parts help each other

- The popup projects partial/final recognized text and allows correction before submission.
- SpeechRecognizer delegates to the configured Android recognition provider. Provider availability and privacy are an external boundary.
- A note audio memo records audio for attachment storage; that is not the same operation as speech-to-text.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
VoiceCaptureContent.kt	Shared transcription popup UI, editable draft transcript and capture controls.
OnDeviceTranscriber.kt	Uses the default phone SpeechRecognizer; despite the name, the selected provider may use online recognition.
SpeechServiceDiagnostics.kt	Reports which speech services are available/configured; diagnostic Google package visibility is not bundled Google code.
VoiceVisualization.kt	Renders voice amplitude/motion visuals; this is presentation rather than speech recognition.
VoiceStyle.kt	Voice Style component in voice capture & transcription.
VoiceStyleSettings.kt	Voice Style Settings component in voice capture & transcription.
VoiceRecordDialog.kt	Dialog interface for Voice Record.
AudioMemoManager.kt	Coordinates operations and state for Audio Memo.

> **When changing this:** The class name OnDeviceTranscriber does not guarantee local recognition. It calls the default Android recognizer; the provider can use its own network permission even though RUN has no INTERNET permission.

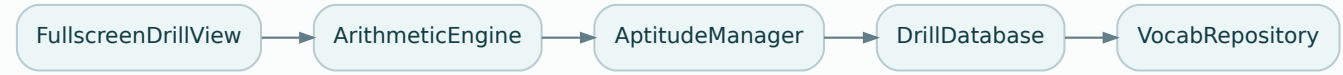
Material / infrastructure: Android SpeechRecognizer/provider; no bundled Google speech SDK established by the dependency list.

Arithmetic, drills & vocabulary

Keep learning sessions, progress and dictionary content connected without duplicating the dictionary.

What you see

Aptitude practice, drill questions/solutions/summary, vocabulary study and saved dictionary words.



How the parts help each other

- Question generation/evaluation is separate from question presentation and session summaries.
- The legacy vocab seed and user progress live in VocabDatabase. Wordset's large dictionary remains compressed assets, loaded a few shards at a time.
- Bookmarking a dictionary sense links/imports into native vocabulary; it does not create a note or automatically mark the word learned.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
AptitudeManager.kt	Learning profile, drill statistics and progress persistence.
ArithmeticEngine.kt	Generates/evaluates supported arithmetic learning questions and explanations.
FullscreenDrillView.kt	Fullscreen Drill View component in learning & vocabulary.
FullscreenDrillSummaryView.kt	Fullscreen Drill Summary View component in learning & vocabulary.
FullscreenSolutionsView.kt	Fullscreen Solutions View component in learning & vocabulary.
DrillModels.kt	Drill Models component in learning & vocabulary.
DrillDatabase.kt	App-private Room SQLite drill-session summaries; separate from the encrypted notes/focus stores.
VocabDatabase.kt	Room vocabulary database seeded from vocab.db; distinct from the large Wordset shard dictionary.
VocabRepository.kt	Data access and update boundary for Vocab.
VocabEntities.kt	Vocab Entities component in learning & vocabulary.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Legacy vocab.db has a documented provenance gap. Replacing its bundled content for distribution must migrate by stable IDs without discarding user bookmarks/progress.

Material / infrastructure: Wordset/WordNet with documented notices; project arithmetic logic; legacy 1,773 idioms and 2,027 OWS entries require origin/rights evidence.

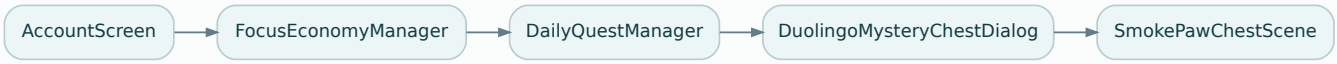
FEATURE 10 / 13

Profile, economy, quests & Pro

Present progress and rewards without letting animation decide the award.

What you see

Profile/avatar/bio, gold, level, activity/streak matrix, achievements, habit telemetry, pinned notes, shortcuts, daily quests, mystery boxes and lifetime Pro.



How the parts help each other

- Profile aggregates several owners: economy, streaks, habits, notes and local settings. It is not one new database of copies.
- DailyQuestManager resolves an award; the chest dialog presents that result. Skip ends presentation, not reward calculation.
- Smoke Paw tier scenes use local vectors through a private offline rasterizer. Lifetime Pro verifies signed tokens locally.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
AccountScreen.kt	User-interface screen for Account.
ProfileAvatarManager.kt	Coordinates operations and state for Profile Avatar.
FocusEconomyManager.kt	Gold/level/avatar economy and profile reward state.
StreakManager.kt	Coordinates operations and state for Streak.
DailyQuestsCard.kt	Daily Quests Card component in profile, rewards & pro.
DailyQuestManager.kt	Quest/chest eligibility and award calculation; visual Skip/Collect must not roll another reward.
MysteryBoxChatCard.kt	Mystery Box Chat Card component in profile, rewards & pro.
DuolingoMysteryChestDialog.kt	Native chest dialog, reveal lifecycle and reward presentation.
SmokePawChestScene.kt	Private offline HTML/vector rasterizer bridge for the approved tiered chest animation.
GamificationHaptics.kt	Gamification Haptics component in profile, rewards & pro.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Do not reroll or award again when replaying/skipping/rotating the animation. Pro uses no declared Play Billing SDK; local paid unlocks alone are not an automatic F-Droid rejection.

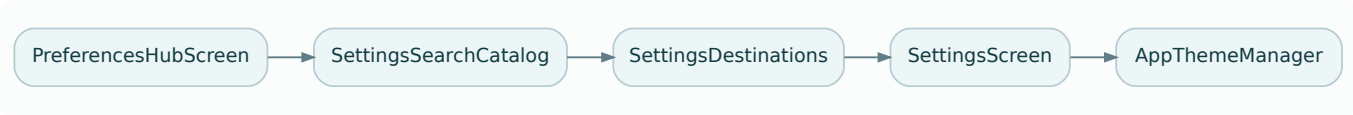
Material / infrastructure: Project economy and local verification; names such as DuolingoComponents do not prove copied upstream code. Chest scene source: assets/smoke_paw_chest.html.

Settings, themes & Vizag scenery

One search/navigation model connects settings categories and shared presentation.

What you see

Appearance, Focus & behavior, Notifications & reminders, Security & permissions, Data & recovery, Bubble settings, Help, diagnostics, private vault and widget editors.



How the parts help each other

- Settings search uses canonical destinations; legacy settings URLs redirect to the relevant category.
- Theme state is shared across app surfaces; widget styling also has its own persisted configuration.
- Nine Vizag variants share a host. Five use the new layered renderer; four retain their older WebP artwork.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
PreferencesHubScreen.kt	User-interface screen for Preferences Hub.
PreferencesHubHelper.kt	Supporting functions for Preferences Hub.
SettingsDestinations.kt	Registers the shared settings routes, search wrappers and legacy-route redirects.
SettingsScreen.kt	User-interface screen for Settings.
SettingsScreenHelper.kt	Supporting functions for Settings Screen.
SettingsSearch.kt	Settings Search component in settings, themes & permissions.
SettingsSearchCatalog.kt	Canonical settings labels/aliases and destinations for in-app search.
SettingsPreferences.kt	Settings Preferences component in settings, themes & permissions.
SettingsPreferenceRules.kt	Settings Preference Rules component in settings, themes & permissions.
AppThemeManager.kt	Coordinates operations and state for App Theme.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Current removed features should stay out of the live map: battery-health widget and Focus DND. Historical screenshots/docs can still mention them.

Material / infrastructure: Five scene pipeline: tools/vizag_replacements/scenes.js and exporter. Retained four: Fishing Harbour, INS Kursura, Vizag Port, Simhachalam. Art inspiration is not proof that site artwork was copied.

Launcher widgets & compact editors

Configure appearance separately from the real task/note records shown by the launcher.

What you see

Tasks, Notes and Exam Countdown widgets. Compact editors use rows, dialogs/tabs, an optional preview and Apply. No battery widget remains.



How the parts help each other

- Configure Activities persist style/selection; Providers build Android RemoteViews, not Compose windows on the launcher.
- Task/note services supply rows from existing repositories. Action receivers handle completion/taps without duplicating the canonical data.
- Countdown event data can be portable; Android widget IDs are local launcher bindings and must be rebound after restore.

Open these files first

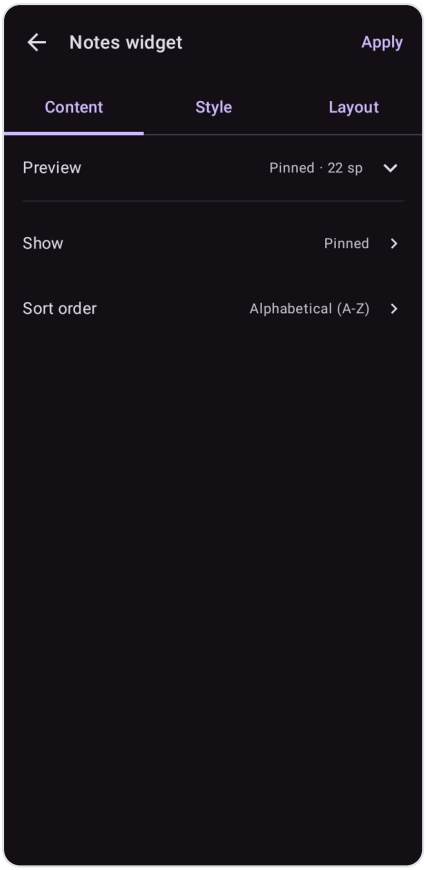
FILE / COMPONENT	WHAT IT OWNS OR HELPS
WidgetEditorComponents.kt	Shared compact customization rows, radio-choice dialogs, sliders, tabs and optional live previews.
TodoWidgetConfigureActivity.kt	Android window / activity entry point for Todo Widget Configure.
WidgetConfig.kt	Configuration models and persistence for Widget.
TodoWidgetProvider.kt	Updates task widget RemoteViews, tabs and row actions.
TodoWidgetService.kt	Todo Widget Service component in home-screen widgets.
TodoWidgetActionReceiver.kt	Handles Android broadcasts or scheduled callbacks for Todo Widget Action.
NoteWidgetConfigureActivity.kt	Android window / activity entry point for Note Widget Configure.
NoteWidgetConfig.kt	Configuration models and persistence for Note Widget.
NoteWidgetProvider.kt	Updates the note widget RemoteViews and attaches actions/content adapters.
NoteWidgetService.kt	Note Widget Service component in home-screen widgets.

Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

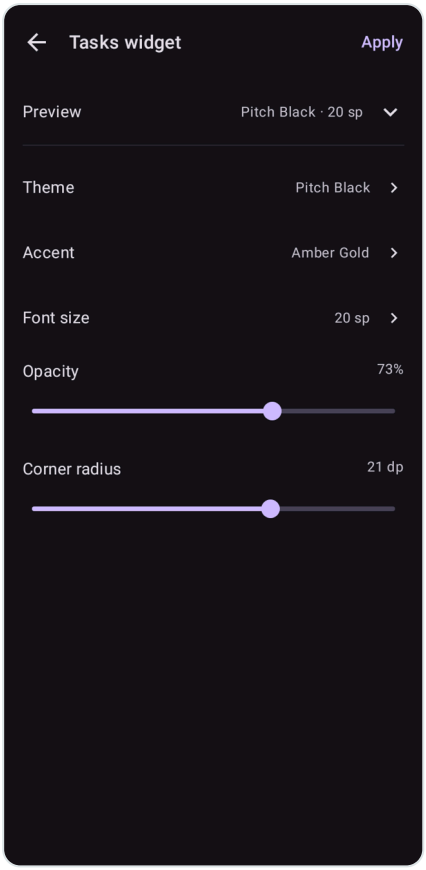
> **When changing this:** Font sizing needs the same explicit typography rules in each RemoteViews path. System font scaling, note content and saved widget font size are different inputs.

Material / infrastructure: Project RemoteViews/configuration components plus Android AppWidget APIs. The six example screenshots are code-rendered fixtures, not redesigned screens.

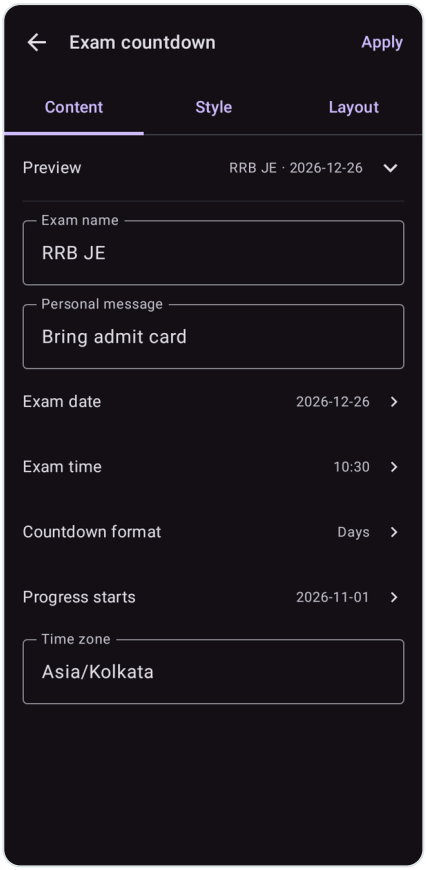
Widget editors · existing UI



Notes · Content



Tasks · Style



Countdown · Content

Current v2.0.37 widget-editor screenshot fixtures; sample user data is synthetic. Actual launcher widgets remain RemoteViews.

FEATURE 13 / 13

Android lifetime, notifications & diagnostics

Keep alarms, services and overlays consistent when the app is hidden or restarted.

What you see

Floating bubble, reminder notifications/overlays, boot recovery, permission checks and diagnostic export.



How the parts help each other

- Boot/app update/time change triggers rebuild eligible reminder schedules; receivers still validate current state before presenting.
- A coordinator serializes task/habit overlay presentation; notification identities are distinct from ongoing foreground services.
- Bubble visibility/snooze and feature enablement are separate states. Hiding in landscape is intentional.

Open these files first

FILE / COMPONENT	WHAT IT OWNS OR HELPS
BootReceiver.kt	Restores optional services and enqueues reminder reconciliation after boot or app update.
PackageChangeReceiver.kt	Handles Android broadcasts or scheduled callbacks for Package Change.
ReminderClockReceiver.kt	Responds to local time/timezone changes by scheduling reminder reconciliation.
ReminderReconcileWorker.kt	Rebuilds reminder schedules, isolating failures; calendar changes preserve absolute task timestamps.
WorkReminderReconcileWorker.kt	Deferred/background work for Work Reminder Reconcile.
BubbleService.kt	Floating chat bubble: foreground lifetime, snooze/visibility, fullscreen/landscape hiding and recovery.
ReminderOverlayStore.kt	Persists/reconstructs pending reminder overlay presentation state.
UnifiedOverlayCoordinator.kt	Queues competing reminder overlays so task/habit windows do not replace one another unpredictably.
TaskReminderOverlayManager.kt	Coordinates operations and state for Task Reminder Overlay.

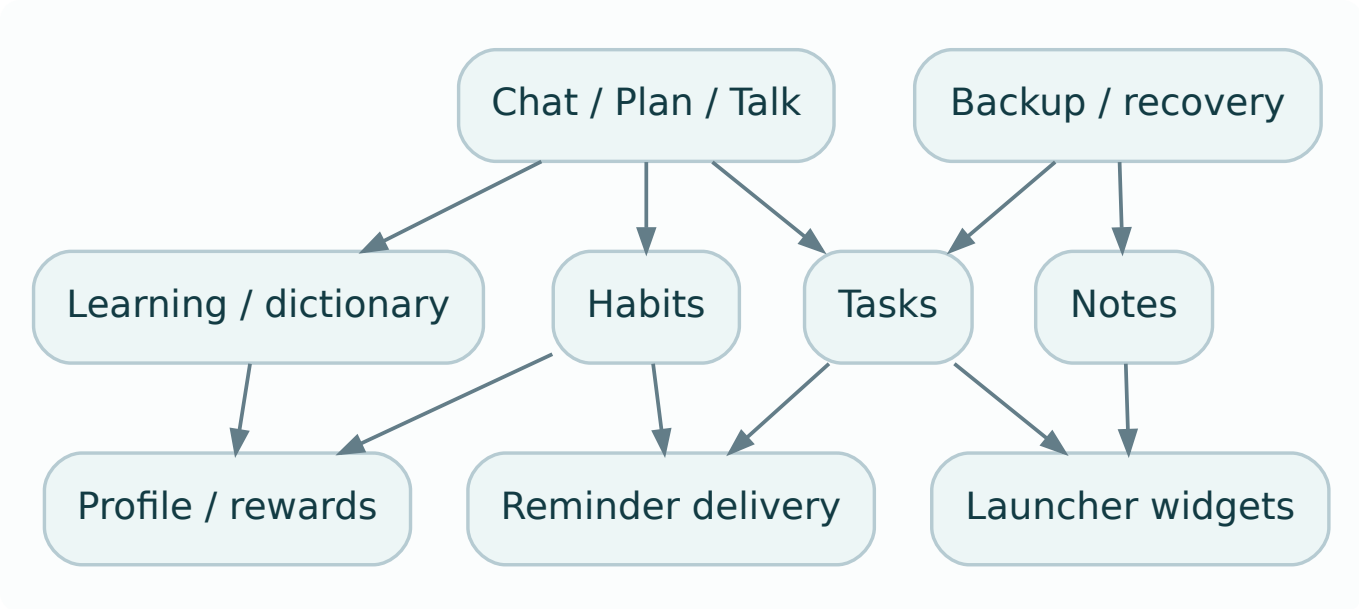
Key files are shown here. The full feature list is in the HTML; every production Kotlin file is described in the code catalog.

> **When changing this:** Android/OEM battery policy and revoked permissions can affect delivery. A foreground service is not a guarantee of exact notification delivery. Do not reactivate a disabled feature from a stale restart alarm.

Material / infrastructure: Project orchestration and Android alarms/WorkManager/WindowManager/Notification APIs. Diagnostics are not evidence of a bundled analytics service.

Follow an action through RUN

These are cross-feature responsibility sequences. They show why changing one small parser, file format or UI action can affect multiple surfaces. They are not packet captures or traces from a live device.



This map highlights selected shared owners and consumers. It is not an exhaustive execution graph.

1. “Plan a Goa trip on the 26th” → a real task

01 →
User text + selected mode

02 →
Date/intent + conversation context

03 →
Draft main task + subtasks

04 →
Edit / remove / summarize / finalize

05 →
Guarded action + receipt

06 →
TaskRepository → FocusDatabase

07 →
List + reminders + widgets

Dates and checklist items belong to a draft until the supported finalization/action path applies them. A summary reads state; it must not become another task. Editing a date must update the same task and reminder schedule. The archive/receipts support revisiting work without creating a parallel task authority.

Trace in: OfflineAssistant.kt · AssistantSession.kt · TripChecklistConversation.kt · PlanTemporalLanguage.kt · AssistantActionExecutor.kt · CanonicalPlanTasks.kt · TaskRepository.kt

2. Add an image → show it in a note

01 → Picker or shared input	**02 →** Import checks + bounded decoding	**03 →** Encrypted attachment file
04 → Note stores media reference	**05 →** EncryptedMediaFetcher decrypts	**06 →** Coil / editor renders image

The image bytes and note record are separate. Both references and bytes must survive backup/restore. Export deliberately decrypts into a user-requested artifact and needs cleanup; editor display alone should use the encrypted-media fetch path.

Trace in: ShareToNoteActivity.kt · NoteImageHelper.kt · NoteMediaManager.kt · EncryptedMediaStorage.kt · NoteEntity.kt · EncryptedMediaFetcher.kt

3. Restore a backup → recover safely after interruption

01 → Pick file + enter password	**02 →** Parse bounds / authenticate / validate	**03 →** Stage databases + media
04 → Journal + maintenance coordination	**05 →** Apply live state / rebuild references	**06 →** Reload caches / rebind local IDs

An interrupted restore is a recoverable operation, not just a single copy command. Startup checks the journal before managers resume ordinary writes. Database keys, media encryption and launcher widget IDs cannot simply be copied as if all were portable identifiers.

Trace in: CryptoBackupEngine.kt · BackupRestoreManager.kt · RestoreStaging.kt · RestoreJournal.kt · MaintenanceGate.kt · FocusApplication.kt

4. A reminder becomes due while RUN is hidden

01 → Repository’s current record	**02 →** Alarm / WorkManager schedule	**03 →** Receiver revalidates eligibility
04 → Notification + queued overlay	**05 →** User action through shared owner	**06 →** Cancel / reschedule / reconcile

Notifications, overlays and foreground-service indicators have different identities and lifetimes. Pause, deletion, completion or a date change invalidates stale work. Boot/time change rebuilds eligible work; it must not reactivate a disabled bubble feature.

Trace in: TaskReminderHelper.kt · TaskReminderReceiver.kt · TaskNotifications.kt · UnifiedOverlayCoordinator.kt · HabitAlarmScheduler.kt · BootReceiver.kt

5. Look up a word → save a learning sense

01 → Supported lookup intent	**02 →** Normalize / hash word	**03 →** Read matching compressed shard
04 → Select actual sense + credits	**05 →** Bookmark / link vocabulary	**06 →** Native learning quiz/progress

The app does not load a language model or unpack the whole dictionary at startup. Wordset definitions, sense-specific editorial examples and legacy vocabulary are separate sources. Saving a sense does not create a note or mark it learned.

Trace in: AssistantDictionary.kt · AssistantDictionaryStore.kt · AssistantVocabularyFeatures.kt · VocabRepository.kt · VocabDatabase.kt

6. Open a mystery box → collect exactly one award

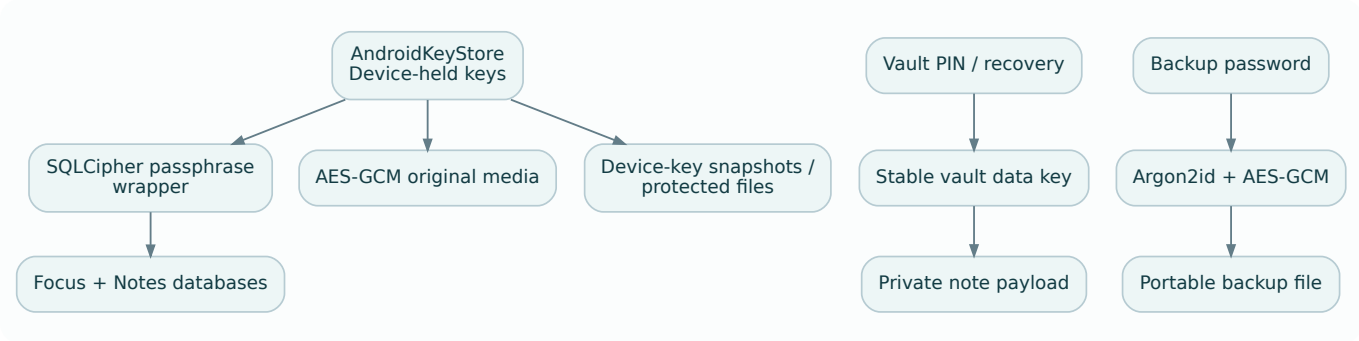
01 → Quest eligibility / completion	**02 →** Resolve reward once	**03 →** Persist reward state
04 → Choose tier’s visual scene	**05 →** Reveal / skip presentation	**06 →** Collect existing result

The animation is presentation. Common, Rare, Epic and Legendary share Smoke Paw artwork with different visual intensity/colors. Skipping, replaying or lifecycle changes must not calculate a new reward.

Trace in: DailyQuestManager.kt · FocusEconomyManager.kt · DuolingoMysteryChestDialog.kt · SmokePawChestScene.kt

Data stores, keys & recovery

There is no single “the app is encrypted” switch. Database encryption, private-note locking, local media protection and portable backups protect different material with different keys.



STORE	FORMAT / PROTECTION	CONTENT AND KEY BOUNDARY
FocusDatabase	SQLCipher-backed Room • schema 16	Tasks, routines, restrictions, habits/logs and assistant receipts **DatabaseKeyProvider wraps the database passphrase using AndroidKeyStore.** Primary productivity store; change migrations and all consumers together.
NoteDatabase	SQLCipher-backed Room • schema 7	Note records, block/text state, labels and media references **DatabaseKeyProvider protects the SQLCipher passphrase; private payload has another vault layer.** Original attachment bytes are separate files, not contained in every note row.
Encrypted attachments	AES-GCM files	Original note media and audio references **Device-held media master key; legacy input handling separate.** Device-bound encryption must be converted appropriately for portable backup.
Private archived payload	Vault subkey / encrypted payload	Private archived note contents **Stable vault data key wrapped with PIN-derived key; recovery flow separate.** Vault PIN is not the entire app lock or portable backup password.
Assistant protected state / snapshots	AYSN1 device-key envelope	Internal snapshots, assistant workspace/session state as used by callers **SnapshotCryptoEngine + AndroidKeyStore-held key.** Device-key files are not a portable recovery plan after uninstall/key loss.

STORE	FORMAT / PROTECTION	CONTENT AND KEY BOUNDARY
AyvaChatDatabase	App-private Room SQLite • schema 1	Chat messages and active drill state **No SQLCipher delegate configured in this database builder.** App-private does not mean world-readable; do not claim extra app-level encryption here.
DrillDatabase	App-private Room SQLite • schema 1	Drill session summaries/progress records **No SQLCipher delegate configured here.** Separate from the encrypted Focus/Notes databases.
VocabDatabase	App-private Room SQLite • schema 3	Legacy seed + user vocabulary/bookmarks/progress **Room seed asset and migrations; no SQLCipher delegate configured here.** Replacing the seed for rights reasons must preserve user learning state.
Preferences & local bindings	Android app-private preferences/files	Theme, settings, widget style/IDs, rotation counters and feature flags **Protection varies by owner; not one universal encrypted preference store.** A settings reset, chat clear and restore have different intended scopes.
Portable encrypted backup	Password envelope • current format 4	Supported exported records/media assembled by BackupRestoreManager **Argon2id 64 MB, t=3, p=1 + AES-256-GCM; legacy readers retained.** Portable with the correct password. Verify actual restore coverage, not just successful decryption.

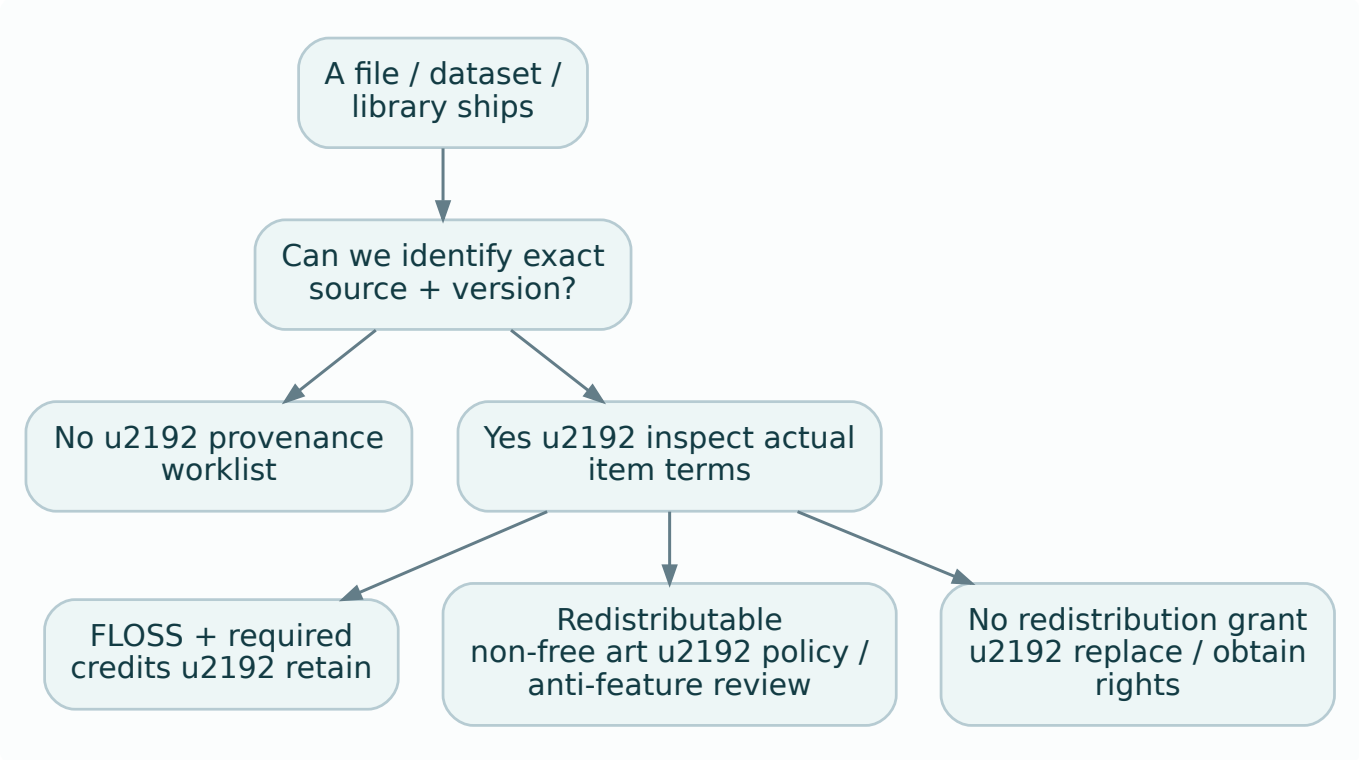
> **Before uninstalling or switching distribution certificates:** make and verify a portable password backup. Device-bound internal snapshots are not enough. KeyStore key availability/hardware backing varies by device, and neither this diagram nor successful decryption proves every restore flow is correct.

Where data leaves a protection boundary

- User-requested PDF/DOCX/share output can be plaintext and needs a scoped temporary-file lease and cleanup.
- The Android speech provider may process audio outside RUN, including on its own network connection.
- A widget exposes only its configured content on the home screen; style config and source note/task records have different owners.
- Diagnostics/backup content must be reviewed for what it contains before sharing; a file being inside the app sandbox is not an encryption claim.

What came from where?

Borrowing can mean code, a dataset, a font, a player library or the art it plays. Each needs its own evidence.



Evidence labels: “Confirmed shipped” means a matching runtime asset/build path is present. “Project-declared” is documentation or an ownership statement. “Unknown” means rights/source evidence is missing. “Research only” is not evidence of incorporation. These labels do not accuse anyone of copying.

Confirmed shipped data

Wordset dictionary

`app/src/main/assets/dictionary/ • tools/import_dictionary.py`

Evidence: Wordset commit 4503bfafbe4556361a296b9d368c18168dfb56c3; docs/assistant/DICTIONARY_DATA.md

Terms / limit: CC BY-SA 4.0 plus retained WordNet notice. Preserve contributors, links and change labels; adapted data stays under the same license.

Practical next step: Keep; complete data/source/credits chain already documented.

Confirmed shipped native component

ChatScript / WordNet / LLVM

`jniLibs/*/librun_dialogue.so • assets/conversation/native-v1.* • tools/experiments/chatscript/`

Evidence: ChatScript commit 9f5eec4736ba22bd992a6498c1e0052e2a795125; source patch, worker and packaging scripts; bundled notices

Terms / limit: ChatScript MIT; component WordNet/LLVM notices. App-authored rule pack has its own project provenance.

Practical next step: Keep functionality; add a source-build recipe rather than relying on checked-in .so or binary archive.

Known provenance gap

Legacy vocabulary seed

`app/src/main/assets/vocab.db`

Evidence: DICTIONARY_DATA.md explicitly records missing source documentation for 1,773 idioms and 2,027 OWS entries.

Terms / limit: No separate license established. Wordset’s license does not cover this seed.

Practical next step: Recover source/permission or replace the bundled seed; preserve user learning data through migration.

Downloaded art; item terms unknown

20 dotLottie files + raw cat JSON

`app/src/main/assets/*.lottie • res/raw/cat_animation.json`

Evidence: Creator/download URL and license version generally absent. Generic LottieFiles manifest author is not item-level evidence.

Terms / limit: Lottie runtime Apache 2.0 is separate. Simple License permits redistribution subject to its conditions, including an asset-service restriction; exact item/date terms must be matched.

Practical next step: Recover each page/license; if redistributable but not FLOSS, discuss NonFreeAssets with F-Droid or replace art.

Downloaded art; item terms unknown

6 Rive cat files

`app/src/main/assets/*.riv`

Evidence: Original item/creator records need recovery.

Terms / limit: Current public Rive Community Content is CC BY 4.0 under Rive terms C.6; private/premium items can differ. Runtime MIT is separate.

Practical next step: Verify qualifying public source; provide creator, source, license and modification credits, or replace unresolved items.

Embedded license evidence

Bundled fonts

`app/src/main/res/font/*.ttf`

Evidence: Name-table copyright and license fields extracted in the file inspector. Allura OFL additionally bundled.

Terms / limit: Caveat, Nunito, Roboto Mono, Roboto Serif, Allura: OFL markers. Roboto Slab: Apache 2.0 marker. Match actual file and retain full required notices.

Practical next step: Keep when evidence/notice chain complete; enumerate authors and license text in third-party credits.

Project source and artwork pipeline

Vizag scenes / Smoke Paw chest

`tools/vizag_replacements/` • `res/raw/vizag_*` • `assets/smoke_paw_chest.html`

Evidence: Approved source image, scene paths and export code; chest design notes. Bare Browser was an inspiration in the conversation.

Terms / limit: Project-wide AGPL declaration is not an independent audit of every artistic contribution. Source/ownership should be recorded.

Practical next step: Keep exact current art; document authorship, generation/edit history and rights for distribution. No proof established of copied Bare Browser SVG.

Project content; verify per item

Stories, facts, games, quotations

`ExperienceContent.kt` • `EntertainmentPack.kt` • `ConversationStories.kt` • `FocusQuotes.kt` • `SummaryQuotes.kt`

Evidence: Docs identify original retellings and short fact wording with references; traditional plots are not a license for a modern retelling.

Terms / limit: Do not infer all quotations or knowledge-base text are public domain. Review attribution, source and exact expression.

Practical next step: Retain original retellings; build a per-content source list; replace any copied modern text without adequate rights.

Influence or naming; copying unestablished

Notesnook / Duolingo names

`Notesnook*.kt` • `Duolingo*.kt`

Evidence: A class/file name is insufficient evidence of a copied upstream implementation.

Terms / limit: If a real port is identified, record original file, commit, changes and compatible terms; do not invent attribution from names.

Practical next step: Inspect git history and actual source similarity before categorizing as borrowed code.

Research only unless matched to runtime

Research candidate repositories

`docs/assistant/research/`

Evidence: A reviewed project appearing in research notes is not a dependency declaration or proof of bundled code.

Terms / limit: Apply licenses only to content actually incorporated. GitHub public visibility alone grants no reuse license.

Practical next step: Separate candidate list from shipped-material ledger.

Declared supporting libraries

Gradle libraries & icons

`app/build.gradle.kts • gradle/libs.versions.toml`

Evidence: Pinned direct dependencies; resolved transitive components depend on the actual build.

Terms / limit: Mostly Apache 2.0/MIT/BSD/OFL families; exact version, transitive and NOTICE obligations still matter.

Practical next step: Keep FLOSS libraries; consolidate notices excluded by packaging and verify the final APK.

Project declarations conflict in places

RUN source / name / logo

`LICENSE • README.md • TRADEMARK.md • 135 production Kotlin headers`

Evidence: Root declares AGPLv3; restrictive proprietary/confidential headers persist. Trademark text explicitly calls for F-Droid permission.

Terms / limit: Rights holder must clarify the effective grant and authorize distribution of the name/icon. Trademark rights and source copyright are different.

Practical next step: Reconcile owned-source headers; record F-Droid brand authorization without abandoning the brand.

Animation & font ledger

This is an item-by-item recovery list, not permission to redistribute an unverified file. There are 20 dotLottie archives, 6 Rive files and one separate raw cat JSON. The exact 26 archive/Rive source pages are not established in current evidence.

ASSET	RECORDED ORIGIN / NEXT EVIDENCE NEEDED
cat_action.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_angry.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_awesome_morning.riv	Downloaded Rive file; creator/download page must be recovered separately from MIT runtime license
cat_blunng.riv	Downloaded Rive file; creator/download page must be recovered separately from MIT runtime license
cat_dance.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_dancing.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_error.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_evening.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_flying.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_gold_fish.riv	Downloaded Rive file; creator/download page must be recovered separately from MIT runtime license
cat_googlyeyes.riv	Downloaded Rive file; creator/download page must be recovered separately from MIT runtime license
cat_luna.riv	Downloaded Rive file; creator/download page must be recovered separately from MIT runtime license
cat_morning.lottie	Downloaded animation; exact page/creator missing from current file evidence
cat_wildfairy.riv	Downloaded Rive file; creator/download page must be recovered separately from MIT runtime license
habbitautumcat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitboredcat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitcatplayingbyobservingballs.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitcatssleeping.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitchrismascat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitcutegirlcat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitgojoawseomecat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbithappylaptopcat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbithollowencat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitpopupcat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitscalmcat.lottie	Downloaded animation; exact page/creator missing from current file evidence
habbitsppokycat.lottie	Downloaded animation; exact page/creator missing from current file evidence

ASSET	RECORDED ORIGIN / NEXT EVIDENCE NEEDED
cat_animation.json	Raw JSON: source/creator/terms need verification.

Fonts: evidence embedded in the actual files

FONT	EMBEDDED MARKER / ATTRIBUTION EVIDENCE
caveat.ttf	SIL Open Font License marker; retain authors/notices and verify source file Copyright 2014 The Caveat Project Authors (https://github.com/googlefonts/caveat)
nunito.ttf	SIL Open Font License marker; retain authors/notices and verify source file Copyright 2014 The Nunito Project Authors (https://github.com/googlefonts/nunito)
roboto_mono.ttf	SIL Open Font License marker; retain authors/notices and verify source file Copyright 2015 The Roboto Mono Project Authors (https://github.com/googlefonts/robotomono)
roboto_serif.ttf	SIL Open Font License marker; retain authors/notices and verify source file Copyright 2020 The Roboto Serif Project Authors (https://github.com/googlefonts/RobotoSerif)
roboto_slab.ttf	Apache 2.0 marker; retain authors/notices and verify source file Copyright 2018 The Roboto Slab Project Authors (https://github.com/googlefonts/robotoslab)
vizag_signature.ttf	SIL Open Font License marker; retain authors/notices and verify source file Copyright 2010 The Allura Project Authors (https://github.com/googlefonts/allura)

Keep a durable material ledger

For each asset, store its exact app path + SHA-256, upstream project/item URL, creator, download/source revision, complete license text/version, changes made, required credit and where that credit appears. The supplied asset CSV contains the current evidence and blank fields to complete. A future filename change should not erase its provenance.

What the supporting libraries do

These are the direct declarations parsed from app/build.gradle.kts and gradle/libs.versions.toml. BOM-managed Compose versions resolve during a build. This is not a complete SPDX SBOM of a release APK; final transitive/native dependency verification is still needed.

Shipped runtime declarations

EXACT DECLARATION / VERSION	LICENSE FAMILY AND BOUNDARY
`androidx.core:core-ktx:1.15.0`	Apache 2.0 family; confirm exact notices
`androidx.dynamicanimation:dynamicanimation:1.0.0`	Apache 2.0 family; confirm exact notices
`androidx.lifecycle:lifecycle-runtime-ktx:2.8.7`	Apache 2.0 family; confirm exact notices
`androidx.work:work-runtime-ktx:2.10.0`	Apache 2.0 family; confirm exact notices
`androidx.fragment:fragment-ktx:1.6.2`	Apache 2.0 family; confirm exact notices
`androidx.appcompat:appcompat:1.6.1`	Apache 2.0 family; confirm exact notices
`androidx.compose:compose-bom:2024.09.00`	Apache 2.0 family; confirm exact notices
`androidx.activity:activity-compose:1.10.1`	Apache 2.0 family; confirm exact notices
`androidx.compose.ui:ui:BOM-managed`	Apache 2.0 family; confirm exact notices
`androidx.compose.ui:ui-graphics:BOM-managed`	Apache 2.0 family; confirm exact notices
`androidx.compose.ui:ui-tooling-preview:BOM-managed`	Apache 2.0 family; confirm exact notices
`androidx.compose.material3:material3:BOM-managed`	Apache 2.0 family; confirm exact notices
`androidx.compose.material:material-icons-core:BOM-managed`	Apache 2.0 family; confirm exact notices
`androidx.compose.material:material-icons-extended:BOM-managed`	Apache 2.0 family; confirm exact notices
`androidx.lifecycle:lifecycle-runtime-compose:2.8.7`	Apache 2.0 family; confirm exact notices
`androidx.lifecycle:lifecycle-viewmodel-compose:2.8.7`	Apache 2.0 family; confirm exact notices
`androidx.navigation:navigation-compose:2.8.9`	Apache 2.0 family; confirm exact notices

EXACT DECLARATION / VERSION	LICENSE FAMILY AND BOUNDARY
`org.jetbrains.kotlinx:kotlinx-coroutines-core:1.10.2`	Apache 2.0 family; confirm exact notices
`org.jetbrains.kotlinx:kotlinx-coroutines-android:1.10.2`	Apache 2.0 family; confirm exact notices
`androidx.room:room-runtime:2.6.1`	Apache 2.0 family; confirm exact notices
`androidx.room:room-ktx:2.6.1`	Apache 2.0 family; confirm exact notices
`net.zetetic:android-database-sqlcipher:4.5.4`	SQLCipher Community BSD-style; retain notice
`com.lambdapioneer.argon2kt:argon2kt:1.4.0`	MIT; underlying native notices also relevant
`com.google.zxing:core:3.5.3`	Apache 2.0 family; confirm exact notices
`androidx.camera:camera-camera2:1.3.4`	Apache 2.0 family; confirm exact notices
`androidx.camera:camera-lifecycle:1.3.4`	Apache 2.0 family; confirm exact notices
`androidx.camera:camera-view:1.3.4`	Apache 2.0 family; confirm exact notices
`io.coil-kt:coil-compose:2.7.0`	Apache 2.0 family; confirm exact notices
`androidx.palette:palette-ktx:1.0.0`	Apache 2.0 family; confirm exact notices
`com.airbnb.android:lottie:6.4.0`	Apache 2.0 family; confirm exact notices
`com.airbnb.android:lottie-compose:6.4.0`	Apache 2.0 family; confirm exact notices
`app.rive:rive-android:10.0.1`	MIT runtime; downloaded .riv artwork separate

Build-time code generation

EXACT DECLARATION / VERSION	LICENSE FAMILY AND BOUNDARY
`androidx.room:room-compiler:2.6.1`	Apache 2.0 family; confirm exact notices

Host test dependencies

EXACT DECLARATION / VERSION	LICENSE FAMILY AND BOUNDARY
`junit:junit:4.13.2`	EPL family; verify exact distributed version (test-only here)
`androidx.test:core:1.6.1`	Apache 2.0 family; confirm exact notices
`androidx.test.ext:junit:1.3.0`	Apache 2.0 family; confirm exact notices

EXACT DECLARATION / VERSION	LICENSE FAMILY AND BOUNDARY
`org.jetbrains.kotlinx:kotlinx-coroutines-test:1.10.2`	Apache 2.0 family; confirm exact notices
`org.robolectric:robolectric:4.16.1`	Apache 2.0 family; confirm exact notices
`io.github.takahirom.roborazzi:roborazzi:1.59.0`	Apache 2.0 family; confirm exact notices
`io.github.takahirom.roborazzi:roborazzi-compose:1.59.0`	Apache 2.0 family; confirm exact notices
`io.github.takahirom.roborazzi:roborazzi-junit-rule:1.59.0`	Apache 2.0 family; confirm exact notices
`androidx.compose.ui:ui-test-junit4:BOM-managed`	Apache 2.0 family; confirm exact notices

Device test dependencies

EXACT DECLARATION / VERSION	LICENSE FAMILY AND BOUNDARY
`androidx.test.ext:junit:1.3.0`	Apache 2.0 family; confirm exact notices
`androidx.test:runner:1.6.2`	Apache 2.0 family; confirm exact notices
`androidx.test.espresso:espresso-core:3.7.0`	Apache 2.0 family; confirm exact notices
`androidx.compose:compose-bom:2024.09.00`	Apache 2.0 family; confirm exact notices
`androidx.compose.ui:ui-test-junit4:BOM-managed`	Apache 2.0 family; confirm exact notices

Debug-only tooling

EXACT DECLARATION / VERSION	LICENSE FAMILY AND BOUNDARY
`androidx.compose.ui:ui-tooling:BOM-managed`	Apache 2.0 family; confirm exact notices
`androidx.compose.ui:ui-test-manifest:BOM-managed`	Apache 2.0 family; confirm exact notices

> No obvious Firebase, Google Play Services, ad SDK or Google Play Billing dependency appears in these direct declarations. AndroidX, ZXing and generic Android speech integration are not themselves evidence of a proprietary GMS dependency. That does not replace a full resolved-binary scan.

F-Droid: change what to what

A plausible route to F-Droid exists. The current checkout is not submission-ready with confidence.

The work is mainly evidence, source-build integration, rights consistency and accurate distribution metadata. App blocking, encryption, Android 12+ support and local Pro verification do not by themselves require removal. No application changes were made for this report.

1 — establish rights

01 · Source licensing

Current
Root AGPLv3 alongside 135 production files with explicit proprietary/confidential restrictions.

Required / intended target
A consistent free-software grant for the code actually shipped; compatible third-party grants retained.

Where: LICENSE; README.md; per-file headers in RUN-file-index.csv; imported-source records.

Effect on the app: No feature removal needed. Only the rightsholder can clarify owned-code grants; third-party restrictions cannot be waived by the app owner.

Acceptance evidence: Review all restrictive headers, keep copyright authors, run license checks and resolve exceptions.

1 — establish rights

02 · Name/icon permission

Current
TRADEMARK.md separately requires written permission for F-Droid distribution with RUN branding.

Required / intended target
Explicit authorization covering F-Droid’s permitted builds/distribution, or a deliberately rebranded distribution.

Where: TRADEMARK.md and F-Droid submission discussion.

Effect on the app: Authorization can keep RUN branding and applicationId. Fork-renaming rules need not be applied to an authorized official distribution.

Acceptance evidence: Ensure the authorization is public, unambiguous and compatible with the planned signing/build arrangement.

1 — establish rights

03 · Downloaded art & legacy text

Current
26 animation files lack a complete item source/terms ledger; legacy vocab.db provenance missing. Raw cat JSON, quotes, images and knowledge text need a rights sweep.

Required / intended target
For each item: author + original URL + license/version + changes + required notices. Replace only unresolved/non-redistributable items.

Where: assets/*.lottie; assets/*.riv; assets/vocab.db; res/raw; content Kotlin/JSON; asset ledger.

Effect on the app: Do not delete user notes or vocabulary progress. Alternative art can preserve the feature; seed replacement needs stable-ID migration.

Acceptance evidence: Match the actual binary/data to source evidence. Generic “free download” or engine license is insufficient.

2 — reproducible source

04 · Native dialogue build

Current
Four checked-in librun_dialogue.so files plus generated native-v1.zip; Gradle does not rebuild them from the pinned upstream source.

Required / intended target
F-Droid metadata builds the supported native worker from pinned ChatScript + tracked patch/wrapper and recreates rule artifacts/hashes.

Where:
tools/experiments/chatscript/{build_android.py,restricted.patch,worker.cpp,package_runtime.py}; app/src/main/jniLibs; assets/conversation.

Effect on the app: Keep native dialogue and its Kotlin fallback. Build coverage must match the ABIs promised by the distribution; non-shipping placeholders need explicit handling.

Acceptance evidence: Clean checkout without prebuilt .so; verify pinned source/NDK; run smoke/worker checks; inspect APK; use scanner and recipe review. Do not blindly ignore binaries.

2 — notices

05 · Third-party credit delivery

Current
Gradle excludes several LICENSE/NOTICE resources. Some notices are bundled, but an exact final-APK notice inventory is incomplete.

Required / intended target
Consolidated third-party notices/assets credits reachable in the app and retained in source/artifacts as required.

Where: app/build.gradle.kts packaging resources; assets/licenses; dictionary/LICENSE.txt; Help/about.

Effect on the app: Small text addition; no need to replace good libraries merely to remove notice obligations.

Acceptance evidence: Compare final dependency graph and bundled assets with notices; include SQLCipher, Argon2Kt, Rive/Lottie, fonts and transitive requirements.

3 — build & metadata

06 · F-Droid recipe

Current
GitHub Actions builds/signs a single Android 12+ release APK; no F-Droid recipe demonstrated.

Required / intended target
An accepted fdroiddata recipe with source revision, Gradle/SDK/NDK settings, binary handling and accurate versions/metadata.

Where: New metadata/com.focusbyrj.app.yml in fdroiddata; app/build.gradle.kts; .github/workflows; source-generation scripts.

Effect on the app: Existing GitHub release can remain. Normal FLOSS Maven dependencies are not equivalent to unexplained checked-in executable blobs.

Acceptance evidence: Run fdroid scanner/build in the F-Droid environment. This guide has not run an official F-Droid build and cannot guarantee approval.

3 — build & metadata

07 · Privacy & feature descriptions

Current
No RUN INTERNET permission; SpeechRecognizer uses the default provider. Fastlane still mentions battery-health statistics after removal.

Required / intended target
Describe core offline operation, optional provider-dependent speech, real permission reasons and current features accurately.

Where: OnDeviceTranscriber.kt; SpeechServiceDiagnostics.kt; AndroidManifest.xml; README.md; fastlane/metadata/android/en-US/full_description.txt.

Effect on the app: Keep voice as optional provider integration. An offline-capable provider may be chosen; do not silently promise offline recognition on all phones.

Acceptance evidence: Check merged manifest, provider/network behavior and storefront text. Declare applicable anti-features honestly with maintainers.

3 — distribution decision

08 · Art anti-feature

Current
Some art may permit redistribution while imposing conditions that are not fully free-software compatible.

Required / intended target
Use FLOSS art, or if policy permits the exact redistributable non-free art, mark NonFreeAssets; no non-free runtime code.

Where: Animation/license ledger; source credits; F-Droid metadata.

Effect on the app: NonFreeAssets is distinct from non-free SDKs. Current policy allows some redistributable non-free assets; not all asset uncertainty means rejection.

Acceptance evidence: Show actual item license to maintainers. Rive public CC BY 4.0 and Lottie Simple License are different cases.

4 — protect users

09 · Signing & migration

Current
Existing installs use the GitHub release certificate. A different signing key cannot replace that install in place.

Required / intended target
Choose reproducible upstream-signature verification/copying if accepted, or plan a safe explicit distribution migration.

Where: Release signing/build settings; F-Droid reproducibility metadata; portable backup/restore UX.

Effect on the app: Do not share private signing keys. Avoid instructing users to uninstall before verified portable backup; device-key snapshots do not survive key loss.

Acceptance evidence: Install/update on a separate test device; test password backup restore, notes/media/vault, tasks and widget rebind. Document certificate ownership.

4 — submission

10 · Eligibility & review

Current
Initial source history is 19 Sep 2026; assessment is 3 Nov 2026. Public maturity is under the usual three-month guideline.

Required / intended target
Demonstrate the app has users/qualifies for the documented exception, or meet normal maturity; submit complete evidence.

Where: Public repository history; submission issue; build metadata and screenshots.

Effect on the app: No need to weaken encryption, remove app blocking, drop Android 12+, or automatically remove local Pro unlocking.

Acceptance evidence: Maintainers decide inclusion. Policy, scanners and dependency signatures can change after this dated snapshot.

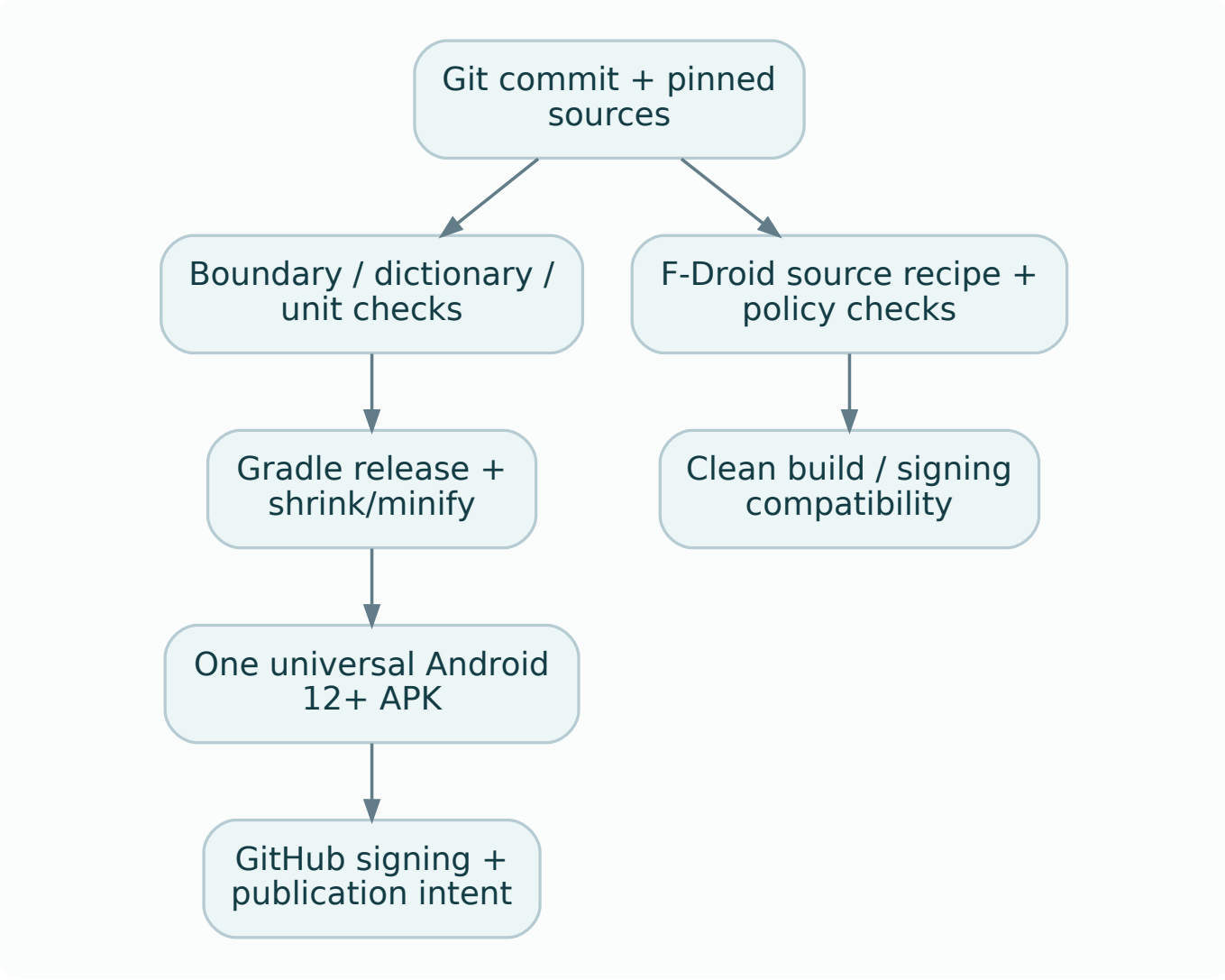
A safe order

- Resolve code/brand rights and material provenance first; classify actual assets rather than removing whole features.
- Build the native worker/rule pack from pinned source and consolidate exact notices.
- Prepare the F-Droid recipe, accurate metadata/anti-features and signing strategy.
- Prove clean builds and safe update/backup/restore on test devices, then submit for maintainer review.

> F-Droid makes the final inclusion decision. This is a dated code/policy assessment, not an official acceptance review, successful F-Droid build or legal opinion. A NonFreeAssets declaration does not cure missing redistribution permission.

Build, release & distribution

The current GitHub release path produces one universal APK for Android 12+; CPU-specific native libraries inside it do not mean separate apps. Compile/target SDK is 35 and minSdk is 31 in this checkout.



FILE	PURPOSE
`build.gradle.kts`	Android version, min/target SDK, shrinking, packaging, dependencies and conditional signing.
`libs.versions.toml`	Library/plugin version catalog.
`validate.yml`	Runs boundary/dictionary checks, unit tests and unsigned builds on changes/PRs.
`release.yml`	Builds/tests, conditionally signs, prepares one universal APK, publishes only when release intent permits.
`release_intent.py`	Distinguishes an intended release/retry from a change that should not publish a duplicate version.
`package_release_apks.py`	Checks that release output is a single universal APK, names it and writes checksums/manifest.
`proguard-rules.pro`	Release shrinker keep rules for reflection, serialization and Android/native integration.

FILE	PURPOSE
settings.gradle.kts	Repository/plugin/dependency-resolution setup.

Three separate questions

- Does it compile and pass the chosen checks? That is build correctness.
- Can another builder recreate the eligible artifact from approved source? That is the source/reproducibility question.
- Can that artifact update an existing install with its certificate? That is Android signing compatibility.

Before you change something

Use this as a starting checklist before changing a feature. It shows consumers worth inspecting; it is not a demand to rewrite working code.

IF YOU CHANGE…	START IN…	ALSO VERIFY…
Bubble icon / shared app artwork	BubbleIconManager.kt • BubbleIconPicker.kt • BubbleService.kt	Use existing app icons, retain selected ID, refresh an existing overlay, preserve snooze/disabled state, and match the docking preview.
Bottom Profile destination	MainActivity.kt • FocusBottomBar.kt • Screen.kt	Dock tap, back stack, top avatar → Settings, return from settings.
Task or plan date behavior	SmartDateParser.kt + caller in assistant/task UI + TaskRepository	Same-turn date, later correction, summary, removal, recurrence, reminder and widget agreement.
Note schema / block serialization	NoteEntity.kt • NoteDatabase.kt • RichTextEngine.kt • NotesnookBlockModel.kt	Old notes, tables/checklists, media references, export, backup and widget display.
Media import/export	NoteImageHelper.kt • NoteMediaManager.kt • EncryptedMediaStorage.kt • ArticleExporter.kt	Oversized/corrupt input, cancellation, temp leases, private-vault display, backup byte integrity.
Encryption or restore	CryptoBackupEngine.kt • BackupRestoreManager.kt • RestoreJournal.kt • DatabaseKeyProvider.kt	Wrong password, tampering, legacy formats, interrupted commit, restart recovery and data retention.
Habit pause	HabitPause.kt • HabitRepository.kt • HabitAlarmScheduler.kt • assistant habit editing	Intervals, resume, midnight/timezone, paused denominator, shield/streak calculations and stale notifications.
Bubble visibility	BubbleService.kt • BubbleSettingsScreen.kt • BootReceiver.kt	Disable vs snooze, ten-minute return when enabled, landscape hide, screen off/on and stale restart.
Widget font/editor	NoteWidgetTypography.kt • Provider/Service • config Activities	New/existing widget, note switch, resize, launcher recreation, font scale and Apply persistence.
Game/story rotation	ChatExperiences.kt • content files • rotation preferences	Clear chat, restart, category choices, next/hint/answer, no repeated fixed first item.
New downloaded artwork	assets/res + renderer + provenance ledger	Exact source/creator/license, app credit, reduced motion, light/dark, lifecycle and byte budget.
Native dialogue version	build_android.py • restricted.patch • package_runtime.py • NativeGeneralDialogue.kt	Pinned source/ABI, restricted capability checks, crash fallback, zip digest and source-build reproducibility.
Settings or storefront copy	SettingsSearchCatalog.kt • SettingsDestinations.kt • fastlane/metadata	Search aliases, sensitive routes, removed battery/DND claims, recognizer privacy and screenshots.

Permission boundaries worth remembering

PERMISSION / CAPABILITY	WHAT IT CONNECTS TO
`SYSTEM_ALERT_WINDOW`	Blocking/reminder/chat overlays; user-controlled special access.
`PACKAGE_USAGE_STATS`	Foreground-app and Screen Time observations; user-granted usage access.

PERMISSION / CAPABILITY	WHAT IT CONNECTS TO
`POST_NOTIFICATIONS`	Reminder/service notification visibility on supported Android versions.
`SCHEDULE_EXACT_ALARM / USE_EXACT_ALARM`	Time-sensitive scheduling declarations/availability; inspect actual manifest and device policy.
`RECORD_AUDIO`	Microphone capture for selected voice/audio features.
`CAMERA`	User-triggered QR/camera flow where available.
`FOREGROUND_SERVICE families`	Declared long-running Android service types and platform restrictions.
`INTERNET`	Explicitly removed from RUN manifest; external speech provider is a separate boundary.

Capabilities above explain feature relationships, not a claim every named scheduling variant is requested. The interactive inspector includes the exact declared permissions and removed entries from AndroidManifest.xml; merged manifest and Android-version policy govern runtime availability.

Tests, fixtures & confidence

The inventory contains 112 tracked files under test roots, including fixtures. Test names describe intended coverage; they do not prove current device behavior or certify security. No application regression suite or official F-Droid build was newly run for this documentation-only task.

AREA	EXAMPLE EXISTING CHECKS
Planning / assistant	TripChecklistConversationTest.kt CanonicalTaskPlanningTest.kt AssistantNotesIsolationTest.kt
Data safety	BackupVerificationTest.kt BackupRestoreHardeningTest.kt RestoreRecoveryDeviceTest.kt
Widgets / typography	WidgetEditorTest.kt NoteWidgetTypographyTest.kt WidgetAuditRegressionTest.kt
Conversation / persistence	ConversationQualitySafetyTest.kt ChatPersistenceConcurrencyTest.kt NativeGeneralDialogueDeviceTest.kt
UI / animation	VoiceAppearanceTest.kt SmokePawChestDeviceTest.kt FocusedDiscoveryDeviceTest.kt

How to use the evidence

- Open the test source and extracted method list in the file inspector to see the scenario and assertions. A filename such as SecurityAuditTest is not a formal security certification.
- Unit/Robolectric checks can exercise rules, serialization and many UI cases. Physical-device checks are still needed for KeyStore, OEM scheduling, speech providers, launchers and overlay lifetime.
- Historical docs and reports are checkpoints. They may mention removed features or old layouts; current code and the pinned revision define this atlas.
- For a future release, run the relevant unit checks, then targeted device flows and source/asset validation. Do not interpret an old green report as validation of a new build.

How this atlas was verified

Every git-tracked file is counted once and hashed. Reviewed feature paths are checked to resolve to real files. Static source references and methods are indexed with their extraction limits. The final HTML is tested without network access; the PDF is checked for readable text, links and representative rendered pages. Validation details are included in atlas-validation.json.

Explore every file

Search a feature, file, class, function or source term.

Click a result to see its role, declarations, methods with source lines, referenced files, reverse references, resources, origin evidence and license status. Filters work on all 1,109 tracked files; ownership groups are explanatory, not a proposed package refactor.

> The interactive browser lives in RUN-Visual-Guide.html. The PDF’s next appendix lists every production Kotlin file; RUN-file-index.csv lists all 1,109 tracked files, including every asset, resource, tool, schema and test. RUN-code-connections.csv contains the 1,878 extracted source edges.

Production code catalog

All 396 production Kotlin files, grouped by explanatory feature owner. Each entry includes its real directory, role and method examples. Click the filename to open the pinned source. Broader function lists and connections are in the HTML inspector.

“Reviewed” marks a file whose role was manually mapped. “Indexed” means the role was inferred from its name/declarations; it is not a line-by-line behavior audit. Some shared helpers are grouped under their primary use.

Find your way around · 12 files

FocusApplication.kt

(app root) · Reviewed

Process startup: restore recovery first, initialize managers, configure encrypted media loading and lazy databases.

Methods: getSharedPreferences(), newImageLoader(), onCreate()

MainActivity.kt

(app root) · Reviewed

Main Android window: wires navigation, primary screens, bottom dock and settings destinations.

Methods: onNewIntent(), handleIncomingIntent(), onCreate(), onResume()

AnalyticsComponents.kt

ui/components · Indexed

Analytics Components component in app shell & navigation.

Methods: HeatmapAndStreaksWidget()

AyvaIcon.kt

ui/components · Indexed

Ayva Icon component in app shell & navigation.

Methods:

DuolingoComponents.kt

ui/components · Indexed

Duolingo Components component in app shell & navigation.

Methods: Duolingo3DButton(), DuolingoShareButton(), DuolingoStatBox(), DuolingoBoltIcon()

FeatureSheet.kt

ui/components · Indexed

Feature Sheet component in app shell & navigation.

Methods:

FocusBottomBar.kt

ui/components · Reviewed

Five-slot bottom dock. Its fourth slot is a blank placeholder; Habits and Routines are reached elsewhere.

Methods: FocusBottomBar()

ProfessionalSlider.kt

ui/components · Indexed

Professional Slider component in app shell & navigation.

Methods: ProfessionalSlider()

Screen.kt

ui/navigation · Reviewed

Route-name registry. The route named empty opens Notes; empty_tab is the blank placeholder.

Methods:

SettingsDestinations.kt

ui/navigation · Reviewed

Registers the shared settings routes, search wrappers and legacy-route redirects.

Methods: settingsDestinations()

ImageUtils.kt

util · Indexed

Image Utils component in app shell & navigation.

Methods: clearCache(), getAppIcon(), drawableToImageBitmap()

SummaryQuotes.kt

util · Indexed

Summary Quotes component in app shell & navigation.

Methods: getNextMorningQuote(), getNextEveningQuote(), getNextQuoteFromDeck(), createNewShuffledDeck()

Focus, app blocking & routines · 31 files

AppRepository.kt

data · Indexed

Data access and update boundary for App.

Methods: getRestriction(), toggleRestriction(), updateMode(), saveApp()

AppRestriction.kt

data · Indexed

App Restriction component in focus, blocking & routines.

Methods:

AppRestrictionDao.kt

data · Indexed

Database query interface for App Restriction.

Methods: getAllRestrictions(), getAllRestrictionsSync(), getRestriction(), insertRestriction()

FocusDatabase.kt

data · Reviewed

Room schema for restrictions, routines, tasks, habits/logs and assistant receipts; encrypted via FocusApplication.

Methods: migrate(), migrate(), migrate(), migrate()

FocusDatabaseMigrationHelper.kt

data · Indexed

Supporting functions for Focus Database Migration.

Methods: getEncryptedDatabaseName(), quarantineFallbackDatabase(), isPlaintextSqliteFile(), tableExists()

FocusSchedule.kt

data · Indexed

Focus Schedule component in focus, blocking & routines.

Methods: isActiveAt()

ScheduleDao.kt

data · Indexed

Database query interface for Schedule.

Methods: getAllSchedules(), getAllSchedulesSync(), getScheduleById(), insertSchedule()

BlockOverlayManager.kt

service · Reviewed

Builds hard/soft blocking windows using WindowManager overlays; this is the primary blocker presentation.

Methods: showBlockScreen(), launchBlockActivity(), tryShowOverlay(), dispatchKeyEvent()

FocusBlockerService.kt

service · Reviewed

Monitors foreground app usage and focus/routine state; decides when direct blocking overlays are needed.

Methods: setOnExitListener(), notifyExited(), isExitSuppressed(), onNewForegroundAppDetected()

FocusDeviceAdminReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Focus Device Admin.

Methods:

FocusSessionEndReceiver.kt

service · Reviewed

Receives the focus-end alarm and completes the timer outside the visible screen.

Methods: onReceive()

AppModeSection.kt

ui/components · Indexed

App Mode Section component in focus, blocking & routines.

Methods: sanitizeAppsForDisplay(), AppModeDropZone()

BoundaryControls.kt

ui/components · Indexed

Boundary Controls component in focus, blocking & routines.

Methods: matches(), toggle(), finish(), BoundaryControls()

CustomRestrictionSection.kt

ui/components · Indexed

Custom Restriction Section component in focus, blocking & routines.

Methods: CustomRestrictionSection(), CompactRestrictionBox()

AddRestrictionScreen.kt

ui/screens · Indexed

User-interface screen for Add Restriction.

Methods: getCategoryForApp(), AddRestrictionScreen(), ModeSelector()

BlockActivity.kt

ui/screens · Reviewed

Manifest-declared alternate blocking Activity. Do not confuse it with the primary direct-overlay service path.

Methods: onReceive(), onCreate(), goHome(), onDestroy()

CategoryEditorHelper.kt

ui/screens · Indexed

Supporting functions for Category Editor.

Methods: filterAndDeduplicateApps()

CustomCategoryEditor.kt

ui/screens · Indexed

Custom Category Editor component in focus, blocking & routines.

Methods: CustomCategoryEditor()

DashboardScreen.kt

ui/screens · Indexed

User-interface screen for Dashboard.

Methods: DashboardScreen(), NormalDashboard(), DeepWorkCard(), ActiveSessionScreen()

SchedulesScreen.kt

ui/screens · Indexed

User-interface screen for Schedules.

Methods: SchedulesScreen(), RoutineCard(), CreateRoutineScreen(), TimePickerBox()

TimeScreen.kt

ui/screens · Indexed

User-interface screen for Time.

Methods: TimeScreen(), loadStats(), AppUsageItem()

FocusViewModel.kt

ui/viewmodels · Indexed

Reactive screen state and operations for Focus.

Methods: triggerNavigation(), clearNavigation(), triggerOpenAddDialog(), clearOpenAddDialog()

AppCatalog.kt

util · Indexed

App Catalog component in focus, blocking & routines.

Methods: namespace(), category(), protectedPackages(), requireEligible()

CustomCategoryManager.kt

util · Indexed

Coordinates operations and state for Custom Category.

Methods: init(), getCategories(), saveCategory(), deleteCategory()

FocusQuotes.kt

util · Indexed

Focus Quotes component in focus, blocking & routines.

Methods: getQuoteOrDefault()

FocusSessionManager.kt

util · Reviewed

Restores and tracks the focus timer deadline; timer completion does not control phone-wide DND.

Methods: finish(), bootCount(), endIntent(), scheduleEnd()

FocusStatsManager.kt

util · Indexed

Coordinates operations and state for Focus Stats.

Methods: fromId(), init(), addInterception(), setHeatmapTheme()

StreakManager.kt

util · Indexed

Coordinates operations and state for Streak.

Methods: fromId(), init(), setStreakSource(), getActiveCurrentStreak()

TemporaryUnlockManager.kt

util · Indexed

Coordinates operations and state for Temporary Unlock.

Methods: grantUnlock(), isUnlocked(), revokeUnlock(), pruneExpiredUnlocks()

UsageBreakTracker.kt

util · Indexed

Usage Break Tracker component in focus, blocking & routines.

Methods: onScreenOff(), onScreenOn(), onForegroundPackageChecked(), sendBubbleReminder()

UsageStatsHelper.kt

util · Indexed

Supporting functions for Usage Stats.

Methods: isSystemOrLauncherPackage(), hasUsageStatsPermission(), requestUsageStatsPermission(), getTodayUsageMap()

Tasks, dates & reminders · 26 files

ProductivityLibrary.kt

data · Indexed

Productivity Library component in tasks, dates & work sessions.

Methods: entries(), put(), delete(), validate()

RecurringTaskWork.kt

data · Indexed

Recurring Task Work component in tasks, dates & work sessions.

Methods: forNextOccurrence(), localDay()

Task.kt

data · Indexed

Task component in tasks, dates & work sessions.

Methods: toJsonObject(), listToJson(), listFromJson()

TaskCompletionRules.kt

data · Indexed

Task Completion Rules component in tasks, dates & work sessions.

Methods: validate()

TaskDao.kt

data · Indexed

Database query interface for Task.

Methods: getAllTasks(), getTrashedTasks(), getTrashedTasksSync(), insertTask()

TaskPlanning.kt

data · Indexed

Task Planning component in tasks, dates & work sessions.

Methods: withSessions(), validate(), json(), remap()

TaskRepository.kt

data · Indexed

Data access and update boundary for Task.

Methods: insertTask(), updateTask(), deleteTask(), moveToTrash()

WorkSession.kt

data · Indexed

Work Session component in tasks, dates & work sessions.

Methods: validate(), json(), read(), readList()

TaskActionReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Task Action.

Methods: onReceive()

TaskReminderOverlayManager.kt

service · Indexed

Coordinates operations and state for Task Reminder Overlay.

Methods: showReminderOverlay(), showReminderOverlayDirect(), tryShowOverlay(), dp()

TaskReminderReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Task Reminder.

Methods: onReceive()

WorkReminderReconcileWorker.kt

service · Indexed

Deferred/background work for Work Reminder Reconcile.

Methods: doWork()

WorkSessionReminderReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Work Session Reminder.

Methods: onReceive()

ProductivityLibraryUi.kt

ui/components · Indexed

Productivity Library Ui component in tasks, dates & work sessions.

Methods:

TaskDependencyEditor.kt

ui/components · Indexed

Task Dependency Editor component in tasks, dates & work sessions.

Methods:

TaskPlanningEditor.kt

ui/components · Indexed

Task Planning Editor component in tasks, dates & work sessions.

Methods: TaskPlanningEditor(), change(), pick()

TaskSpeedDialFab.kt

ui/components · Indexed

Task Speed Dial Fab component in tasks, dates & work sessions.

Methods: TaskSpeedDialFab(), SpeedDialOption()

QuickAddTaskActivity.kt

ui/screens · Indexed

Android window / activity entry point for Quick Add Task.

Methods: onCreate(), QuickAddTaskOverlay()

TaskReminderPopupActivity.kt

ui/screens · Indexed

Android window / activity entry point for Task Reminder Popup.

Methods: onCreate(), TaskReminderPopupScreen(), RescheduleChip()

TodosScreen.kt

ui/screens · Indexed

User-interface screen for Todos.

Methods: TodosScreen(), TodoSegmentedPill(), TaskItem(), CompletedAccordion()

TaskViewModel.kt

ui/viewmodels · Indexed

Reactive screen state and operations for Task.

Methods: addTask(), updateTask(), deleteTask(), restoreTask()

CompletedTaskHistoryManager.kt

util · Indexed

Coordinates operations and state for Completed Task History.

Methods: getStartOfTodayMs(), recordCompletedTask(), getTodayCompletedTasks(), getTodayCompletedCount()

SmartDateParser.kt

util · Reviewed

Shared supported natural-language date parsing; role/context decisions remain with the caller.

Methods: isLeapYear(), formatDueDate(), formatDueDateFull(), parse()

TaskNotifications.kt

util · Reviewed

Separates task notification identities from service notifications with full task-ID tags.

Methods: tag(), isChild(), matches(), cancelLegacy()

TaskReminderHelper.kt

util · Reviewed

Schedules/cancels task reminders and updates notification summaries.

Methods: scheduleReminder(), scheduleAllPendingReminders(), cancelReminder(), cancelReminderById()

WorkSessionReminders.kt

util · Indexed

Work Session Reminders component in tasks, dates & work sessions.

Methods: retry(), availability(), file(), read()

Habits & pause/resume · 14 files

Habit.kt

data · Indexed

Habit component in habits & pausing.

Methods: withPaused()

HabitDao.kt

data · Indexed

Database query interface for Habit.

Methods: getAllHabitsSync(), getAllActiveHabits(), getHabitById(), insertHabit()

HabitFrequency.kt

data · Indexed

Habit Frequency component in habits & pausing.

Methods: validate(), json(), scheduled(), weekCount()

HabitPause.kt

data · Reviewed

Pause intervals and pause-aware habit behavior; paused time is not fabricated completion.

Methods: isPausedOn(), day(), contains()

HabitRepository.kt

data · Indexed

Data access and update boundary for Habit.

Methods: getDateFormat(), getTodayDateString(), getAllActiveHabits(), getTodayLogs()

HabitActionReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Habit Action.

Methods: onReceive(), cleanUpHabitSummary()

HabitFloatingOverlayManager.kt

service · Indexed

Coordinates operations and state for Habit Floating Overlay.

Methods: getRandomHabitLottieAsset(), showHabitOverlay(), showHabitOverlayDirect(), onViewAttachedToWindow()

HabitReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Habit.

Methods: getNotificationId(), createHabitNotificationChannel(), getActionLabelForHabit(), onReceive()

HabitFrequencyEditor.kt

ui/components · Indexed

Habit Frequency Editor component in habits & pausing.

Methods:

HabitsChatCard.kt

ui/screens · Indexed

Habits Chat Card component in habits & pausing.

Methods: HabitsChatCard(), HabitsListView(), EnterHabitView(), autoSplitInterval()

HabitsScreen.kt

ui/screens · Indexed

User-interface screen for Habits.

Methods: HabitsScreen(), HabitProgressHeroCard(), HabitPresetsRow(), HabitCard()

HabitViewModel.kt

ui/viewmodels · Indexed

Reactive screen state and operations for Habit.

Methods: addHabit(), updateHabit(), setPaused(), deleteHabit()

HabitAlarmScheduler.kt

util · Reviewed

Computes and registers habit reminder windows; pause/resume affects scheduling.

Methods: getPendingIntent(), getRequestCode(), scheduleHabitReminder(), snoozeHabit()

HabitMicroCopyProvider.kt

util · Indexed

Provider / update entry point for Habit Micro Copy.

Methods: getStreakBadge(), getCategory(), getCurrentTimeOfDay(), getProgressState()

Notes, editor & attachments · 42 files

NoteDao.kt

data/note · Indexed

Database query interface for Note.

Methods: getAllActiveNotes(), getArchivedNotes(), getTrashedNotes(), getNoteById()

NoteDatabase.kt

data/note · Reviewed

SQLCipher-backed note database with explicit migration/recovery handling.

Methods: migrate(), migrate(), migrate(), migrate()

NoteDatabaseMigrationHelper.kt

data/note · Indexed

Supporting functions for Note Database Migration.

Methods: getEncryptedDatabaseName(), checkAndMigrateIfLegacyPlaintextExists(), secureWipeAndDelete(), isPlaintextSqliteFile()

NoteEntity.kt

data/note · Reviewed

Stored note model, checklist/media references, privacy flags and editor content fields.

Methods: getImageUris(), getAudioUris(), getChecklistItems(), getLabels()

NoteImageHelper.kt

data/note · Reviewed

Image import/decoding helpers; image safety and dimensions are separate from editor styling.

Methods: processAndSaveImage(), processAndSaveBitmap(), rotateBitmap(), flipBitmap()

NoteMediaManager.kt

data/note · Reviewed

Imports/manages note attachments and their storage references.

Methods: isAllowedMediaFile(), secureDeleteMediaFile(), secureDeleteMediaFile(), deleteNoteMediaFiles()

NoteRepository.kt

data/note · Indexed

Data access and update boundary for Note.

Methods: getActiveNotes(), getArchivedNotes(), getTrashedNotes(), searchNotes()

NotesStorageVersion.kt

data/note · Indexed

Notes Storage Version component in notes, editor & media.

Methods: current(), replaced()

EditorDrafts.kt

ui/components · Indexed

Editor Drafts component in notes, editor & media.

Methods: beginEditing(), save(), discard(), rememberEditorDraft()

ArticleDocxGenerator.kt

ui/screens/notes · Indexed

Article Docx Generator component in notes, editor & media.

Methods: generateDocx(), addZipEntry(), buildContentTypesXml(), buildRootRelsXml()

ArticleEditorialBottomSheet.kt

ui/screens/notes · Indexed

Article Editorial Bottom Sheet component in notes, editor & media.

Methods: ArticleEditorialBottomSheet()

ArticleExportBottomSheet.kt

ui/screens/notes · Indexed

Article Export Bottom Sheet component in notes, editor & media.

Methods: ArticleExportBottomSheet()

ArticleExporter.kt

ui/screens/notes · Indexed

Article Exporter component in notes, editor & media.

Methods: exportToMarkdown(), exportToMarkdownWithFrontmatter(), convertTextSpansToMarkdown(), exportToHtml()

ArticleExporterHelper.kt

ui/screens/notes · Indexed

Supporting functions for Article Exporter.

Methods: getEffectiveBlocks(), sanitizeFileName(), generateShareFileName(), pruneStaleExportFiles()

ArticlePdfGenerator.kt

ui/screens/notes · Indexed

Article Pdf Generator component in notes, editor & media.

Methods: generatePdf(), ensureSpace(), createStaticLayout()

ArticleTocHelper.kt

ui/screens/notes · Indexed

Supporting functions for Article Toc.

Methods: extractToc(), ArticleTocBottomSheet()

AudioMemoManager.kt

ui/screens/notes · Indexed

Coordinates operations and state for Audio Memo.

Methods: finalizedTranscript(), startRecording(), retryRecording(), pauseRecording()

AudioPlayerWidget.kt

ui/screens/notes · Indexed

Audio Player Widget component in notes, editor & media.

Methods: AudioPlayerCardCompact(), AudioPlayerEditorItem(), calculateSeekPosition()

DocumentStatsDialog.kt

ui/screens/notes · Indexed

Dialog interface for Document Stats.

Methods: calculate(), DocumentStatsBottomSheet(), StatCard(), DetailRow()

EditLabelsDialog.kt

ui/screens/notes · Indexed

Dialog interface for Edit Labels.

Methods: sanitizeLabelsForDisplay(), canAddLabel(), canRenameLabel(), isLabelSelected()

KeepColorDialog.kt

ui/screens/notes · Indexed

Dialog interface for Keep Color.

Methods: KeepColorDialog()

KeepColorPalette.kt

ui/screens/notes · Indexed

Keep Color Palette component in notes, editor & media.

Methods: getBackgroundColor(), getBorderColor(), getTextColor(), resolveBackgroundColor()

KeepFontPalette.kt

ui/screens/notes · Indexed

Keep Font Palette component in notes, editor & media.

Methods: getFont()

KeepImageCollage.kt

ui/screens/notes · Indexed

Keep Image Collage component in notes, editor & media.

Methods: buildGoogleKeepCollageRows(), KeepCardImageCollage(), KeepEditorImageCollage(), EditorImageTile()

KeepNoteCard.kt

ui/screens/notes · Indexed

Keep Note Card component in notes, editor & media.

Methods: KeepNoteCard(), getCleanPreviewContent()

KeepNoteEditor.kt

ui/screens/notes · Reviewed

Main note-editing UI; text, blocks, formatting and attachment entry points.

Methods: KeepNoteEditor(), safeRequestFocus(), syncAndCommitBlocks(), getActiveTextState()

KeepNoteShareParser.kt

ui/screens/notes · Indexed

Parses input into structured data for Keep Note Share.

Methods: parseIntent(), extractImageUris(), isSafeShareUri(), parseContent()

KeepSketchDialog.kt

ui/screens/notes · Indexed

Dialog interface for Keep Sketch.

Methods: calculateLineStep(), coerceColorComponent(), safeRadius(), isCoordinateValid()

KeepThemeIllustration.kt

ui/screens/notes · Indexed

Keep Theme Illustration component in notes, editor & media.

Methods: KeepThemeIllustration(), drawGroceriesTheme(), drawFoodTheme(), drawMusicTheme()

McqTextParser.kt

ui/screens/notes · Indexed

Parses input into structured data for Mcq Text.

Methods: stripMarkdown(), toFormattedText(), resolveAnswer(), toRoman()

NoteLabelsDialog.kt

ui/screens/notes · Indexed

Dialog interface for Note Labels.

Methods: NoteLabelsDialog()

NotesScreen.kt

ui/screens/notes · Indexed

User-interface screen for Notes.

Methods: NotesScreen(), checkAndStartRecording()

NotesSpeedDialFab.kt

ui/screens/notes · Indexed

Notes Speed Dial Fab component in notes, editor & media.

Methods: NotesSpeedDialFab(), SpeedDialOption()

NotesViewModel.kt

ui/screens/notes · Reviewed

Coordinates note screen state, editing, archive/vault actions and repository operations.

Methods: setReturnRouteOnClose(), consumeReturnRoute(), clearVaultKeyMismatch(), refreshVaultStatus()

NotesnookBlockModel.kt

ui/screens/notes · Reviewed

App block-document representation. The filename alone does not prove code was copied from Notesnook.

Methods: serialize(), isArticleMode(), getSubtitle(), parse()

NotesnookBlockWidgets.kt

ui/screens/notes · Indexed

Notesnook Block Widgets component in notes, editor & media.

Methods: NotesnookCodeBlockWidget(), NotesnookCalloutWidget(), NotesnookHorizontalRuleWidget(), NotesnookMathWidget()

NotesnookEditorDrawer.kt

ui/screens/notes · Indexed

Notesnook Editor Drawer component in notes, editor & media.

Methods: NotesnookEditorDrawer(), NotesnookTileButton(), NotesnookListHierarchyPopup(), ListFormatPillColumn()

NotesnookFormattingHelper.kt

ui/screens/notes · Indexed

Supporting functions for Notesnook Formatting.

Methods: applyInlineWrap(), applyLinePrefix(), handleEnterKey(), intToRomanNumeral()

NotesnookInsertBottomSheet.kt

ui/screens/notes · Indexed

Notesnook Insert Bottom Sheet component in notes, editor & media.

Methods: isValidMediaUrl(), NotesnookInsertBottomSheet(), safeDismissAnd(), NotesnookInsertRow()

NotesnookTableWidget.kt

ui/screens/notes · Indexed

Notesnook Table Widget component in notes, editor & media.

Methods: NotesnookTableBuilderDialog(), updateSelection(), NotesnookTableEditorDialog(), ensureGrid()

RichTextEngine.kt

ui/screens/notes · Reviewed

Rich-text transformations and formatting state used by the note editor.

Methods: isValid(), parse(), parseInline(), serialize()

ShareToNoteActivity.kt

ui/screens/notes · Indexed

Android window / activity entry point for Share To Note.

Methods: onCreate(), ShareToNoteOverlay(), saveImportedNote()

Vault, backup & recovery · 29 files

ArchiveVaultSecurity.kt

data/note · Reviewed

Private archived-note PIN/recovery handling; derives wrapping keys and manages a stable vault data key.

Methods: wrapVaultDataKey(), unwrapVaultDataKey(), migrateLegacyNotesCrashSafely(), getActiveVaultSubKey()

DatabaseKeyProvider.kt

data/note · Reviewed

Creates/unwraps the SQLCipher passphrase using AndroidKeyStore protection; it is not the user backup password.

Methods: getOrCreatePassphrase(), wipeByteArray(), clearCachedPassphrase(), getMasterKeyForEncryption()

DataRecoveryTools.kt

ui/components · Indexed

Data Recovery Tools component in encryption, backup & recovery.

Methods: inventory(), chooseBackup(), openPage(), size()

BackupFlow.kt

ui/screens · Indexed

Backup Flow component in encryption, backup & recovery.

Methods: BackupFlow()

DataRecoveryScreen.kt

ui/screens · Indexed

User-interface screen for Data Recovery.

Methods: DataRecoveryScreen()

PrivateVaultSettings.kt

ui/screens · Indexed

Private Vault Settings component in encryption, backup & recovery.

Methods: PrivateVaultSettingsScreen(), PrivateVaultSettingsSheet(), PrivateVaultSettingsPanel(), refresh()

ArchiveVaultDialogs.kt

ui/screens/notes · Indexed

Archive Vault Dialogs component in encryption, backup & recovery.

Methods: ArchiveVaultPinDots(), ArchiveVaultNumericKeypad(), KeypadCircleButton(), ArchiveVaultUnlockDialog()

AutoBackupScheduler.kt

util/backup · Indexed

Auto Backup Scheduler component in encryption, backup & recovery.

Methods: schedule(), cancel(), runNow(), runDailyBackup()

AutoBackupWorker.kt

util/backup · Indexed

Deferred/background work for Auto Backup.

Methods: doWork(), enqueue(), cancel()

BackupRestoreManager.kt

util/backup · Reviewed

Collects/export/restores application data and media; validates/stages before applying live changes.

Methods: recoveryPreferenceNames(), generateBackupFileName(), canCreateBackup(), createEncryptedBackup()

CryptoBackupEngine.kt

util/backup · Reviewed

Portable password-protected backup envelope: current Argon2id + AES-GCM, with legacy readers.

Methods: readFully(), openEncryptingStream(), openDecryptingStream(), encrypt()

DataSafetyManager.kt

util/backup · Reviewed

Schedules and manages internal safety snapshots and recovery evidence.

Methods: writeDailyBackup(), writeDailyBackup(), writePreOpSnapshot(), writePreOpSnapshot()

EncryptedSnapshotFile.kt

util/backup · Indexed

Encrypted Snapshot File component in encryption, backup & recovery.

Methods: write(), migratePlaintext(), read()

ExportCleanupWorker.kt

util/backup · Reviewed

Removes expired temporary plaintext exports; intentional sharing is separate from encrypted original storage.

Methods: doWork(), schedule()

MaintenanceGate.kt

util/backup · Reviewed

Coordinates participating writers during maintenance; not a second database or encryption algorithm.

Methods: inheritedContext(), enter(), exit(), write()

MaintenanceOpenHelperFactory.kt

util/backup · Reviewed

Wraps SQLite access in the maintenance barrier. Without a SQLCipher delegate it does not encrypt a database.

Methods: create(), begin(), beginTransaction(), beginTransactionNonExclusive()

MaintenancePreferences.kt

util/backup · Indexed

Maintenance Preferences component in encryption, backup & recovery.

Methods: edit(), putString(), putStringSet(), putInt()

RestoreJournal.kt

util/backup · Reviewed

Crash-recovery journal: completes or rolls back interrupted restore operations at startup.

Methods: checkpoint(), databases(), journal(), marker()

RestoreStaging.kt

util/backup · Reviewed

Prepares bounded validated restore content before live mutation.

Methods: protectLegacyStages()

SnapshotCryptoEngine.kt

util/backup · Reviewed

Device-key AES-GCM envelope for internal snapshots and protected state files; not a portable password backup.

Methods: getMasterKeyForEncryption(), getMasterKeyForDecryption(), getLegacyFallbackKey(), isEncrypted()

TaskRecoveryStore.kt

util/backup · Indexed

Persists and retrieves state for Task Recovery.

Methods: write(), restore()

VocabularyArchive.kt

util/backup · Indexed

Vocabulary Archive component in encryption, backup & recovery.

Methods: export(), rows(), validate(), restore()

Argon2idKdf.kt

util/crypto · Indexed

Argon2id Kdf component in encryption, backup & recovery.

Methods: isNativeAvailable(), deriveKeyJvmFallback(), deriveKey(), deriveKey()

EncryptedMediaFetcher.kt

util/crypto · Reviewed

Coil bridge for decrypting encrypted note attachments for display.

Methods: fetch(), detectMimeType(), create()

EncryptedMediaStorage.kt

util/crypto · Reviewed

Encrypts original local attachments using AES-GCM with an AndroidKeyStore-held key; legacy reads are distinct.

Methods: getMasterKeyForEncryption(), getMasterKeyForDecryption(), getLegacyFallbackKey(), writeEncryptedBytes()

HkdfUtil.kt

util/crypto · Indexed

Hkdf Util component in encryption, backup & recovery.

Methods: extract(), expand(), deriveKey(), hmacSha256()

VaultPayloadEncryptor.kt

util/crypto · Reviewed

Encrypts/decrypts the private note payload with the active vault subkey.

Methods: isVaultEncrypted(), parseKeyVersion(), stripKeyVersionPrefix(), encryptNotePayload()

VaultCryptoEngine.kt

util/sync · Reviewed

Cryptographic packaging/key handling for the vault-transfer path; distinct from full-app backup envelopes.

Methods: toJsonString(), fromJsonString(), encryptWithRawKey(), decryptWithRawKey()

VaultSyncManager.kt

util/sync · Reviewed

Serializes/restores note/task vault payloads. The name does not establish a running network sync service.

Methods: createVaultJson(), restoreVaultFromJson()

Chat, Plan, Talk & offline experiences · 128 files

AssistantPlanReceipt.kt

data · Indexed

Assistant Plan Receipt component in chat, planning & games.

Methods: get(), insert(), finishEffects(), pending()

ActiveDrillStateEntity.kt

data/chat · Indexed

Active Drill State Entity component in chat, planning & games.

Methods:

AyvaChatDao.kt

data/chat · Indexed

Database query interface for Ayva Chat.

Methods: getAllMessagesFlow(), getAllMessagesSync(), getMessageById(), insertMessage()

AyvaChatDatabase.kt

data/chat · Reviewed

App-private Room SQLite transcript/drill-state database; no SQLCipher delegate is configured here.

Methods: getDatabase()

AyvaChatMessageEntity.kt

data/chat · Indexed

Ayva Chat Message Entity component in chat, planning & games.

Methods: toPersistedChatMessage(), toEntity()

BubbleIconPicker.kt

ui/components · Reviewed

Compact chat-menu picker for the 12 existing app icons; highlights the saved bubble selection and applies on tap.

Methods: BubbleIconPicker()

CatMagicRiveView.kt

ui/components · Indexed

Cat Magic Rive View component in chat, planning & games.

Methods: resolveStateMachine(), CatWelcomeRiveView(), CatMagicRiveView(), updateCoordinates()

TalkActionChips.kt

ui/components · Indexed

Talk Action Chips component in chat, planning & games.

Methods: cleanLabelAndEmoji(), resolveActionColorTheme(), TalkActionChips(), Tactile3DChoiceButton()

BubbleChatActivity.kt

ui/screens · Reviewed

Chat activity and integration shell: composer, message cards, actions and assistant entry point.

Methods: toChatMessage(), toPersistedChatMessage(), createDrillSessionWithQuestions(), onReceive()

AyvaCatPeekingHost.kt

ui/screens/chat · Indexed

Ayva Cat Peeking Host component in chat, planning & games.

Methods: AyvaCatFloatingView(), CatLottiePeekingView(), CatActionLottieView(), MorningBriefLottieHeader()

AyvaChatChrome.kt

ui/screens/chat · Indexed

Ayva Chat Chrome component in chat, planning & games.

Methods: AyvaCompactHeader(), AyvaActiveContext(), AyvaReplyPreview(), AyvaConversationSearch()

AyvaChatTimeline.kt

ui/screens/chat · Indexed

Ayva Chat Timeline component in chat, planning & games.

Methods: parseRichFormattedText(), DuolingoFreezeButton(), PendingActionCard(), TaskSummaryCard()

AyvaComposerBar.kt

ui/screens/chat · Indexed

Ayva Composer Bar component in chat, planning & games.

Methods: filter(), AyvaComposerBar()

AyvaConversationMessage.kt

ui/screens/chat · Indexed

Ayva Conversation Message component in chat, planning & games.

Methods: AyvaConversationMessage(), filteredActions(), AyvaActionReceipt(), AyvaConversationActions()

AyvaPlanningControls.kt

ui/screens/chat · Indexed

Ayva Planning Controls component in chat, planning & games.

Methods: AyvaPlanningControls()

AyvaResponseControls.kt

ui/screens/chat · Indexed

Ayva Response Controls component in chat, planning & games.

Methods: AyvaSelectableTaskList(), AyvaInlineSetting(), preview(), responseActions()

ChatLifecycle.kt

ui/screens/chat · Reviewed

Suspends hidden chat projection and stabilizes viewed timestamps when UI visibility changes.

Methods: rememberChatStateProjection(), toStableChatRecord()

ChatTextSizeDialog.kt

ui/screens/chat · Indexed

Dialog interface for Chat Text Size.

Methods: ChatTextSizeDialog()

AyvaDialogueEngine.kt

util · Indexed

Core transformation or evaluation routines for Ayva Dialogue.

Methods: levenshtein(), isFuzzyMatch(), containsAnyFuzzy(), getNextFromDeck()

AyvaTalkEngine.kt

util · Indexed

Core transformation or evaluation routines for Ayva Talk.

Methods: recordTurn(), getRecentHistory(), stem(), expandTokens()

BubbleChatManager.kt

util · Reviewed

Chat transcript cache and serialized Room persistence, unread counts, startup merge and restore invalidation.

Methods: init(), optNullableString(), parseLegacyJson(), mergeChanges()

BubbleIconManager.kt

util · Reviewed

Shares the existing app icon artwork with the bubble; persists its separate choice and notifies the running overlay without restarting it.

Methods: option(), currentIconRes(), setIcon()

ChatConversationContext.kt

util · Indexed

Chat Conversation Context component in chat, planning & games.

Methods: focusTask(), clear(), resolve()

ChatTaskDraft.kt

util · Indexed

Chat Task Draft component in chat, planning & games.

Methods: parse()

OfflineNluEngine.kt

util · Indexed

Core transformation or evaluation routines for Offline Nlu.

Methods: levenshtein(), matchesAnyFuzzy(), isQuestion(), normalizeRequest()

AdvancedPlanLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Advanced Plan.

Methods: edit(), one(), publish(), change()

AssistantAction.kt

util/assistant · Indexed

Assistant Action component in chat, planning & games.

Methods: toJson(), fromJson(), longs(), encode()

AssistantActionExecutor.kt

util/assistant · Reviewed

Guarded execution owner for assistant-requested changes and undo; response text is not write authority.

Methods: canDisableRoutine(), execute(), now(), undoTask()

AssistantAppControl.kt

util/assistant · Indexed

Assistant App Control component in chat, planning & games.

Methods: inventory(), ids(), text(), resolve()

AssistantAppKnowledge.kt

util/assistant · Indexed

Assistant App Knowledge component in chat, planning & games.

Methods: clean(), contains(), topics(), isWrite()

AssistantCalculator.kt

util/assistant · Indexed

Assistant Calculator component in chat, planning & games.

Methods: answer(), done(), space(), take()

AssistantCapabilities.kt

util/assistant · Indexed

Assistant Capabilities component in chat, planning & games.

Methods: current(), action(), query(), hint()

AssistantChecklist.kt

util/assistant · Indexed

Assistant Checklist component in chat, planning & games.

Methods: validate(), answer()

AssistantCompoundParser.kt

util/assistant · Indexed

Parses input into structured data for Assistant Compound.

Methods: split()

AssistantContext.kt

util/assistant · Indexed

Assistant Context component in chat, planning & games.

Methods: live(), expire()

AssistantConversation.kt

util/assistant · Indexed

Assistant Conversation component in chat, planning & games.

Methods: questions(), saveQuestions(), suspendQuestion(), resumeQuestion()

AssistantConversationSupport.kt

util/assistant · Indexed

Assistant Conversation Support component in chat, planning & games.

Methods: answer(), describe(), tasks()

AssistantDeviceFeatures.kt

util/assistant · Indexed

Assistant Device Features component in chat, planning & games.

Methods: day(), close(), label(), duration()

AssistantDialogue.kt

util/assistant · Indexed

Assistant Dialogue component in chat, planning & games.

Methods: isClockRequest(), reply(), answer(), factor()

AssistantDictionary.kt

util/assistant · Indexed

Assistant Dictionary component in chat, planning & games.

Methods: isLookup(), features(), frame(), plain()

AssistantDictionaryStore.kt

util/assistant · Reviewed

Reads bounded gzip dictionary shards and editorial sense data; not a cloud dictionary.

Methods: removeEldestEntry(), normalize(), relatedForms()

AssistantDiscussion.kt

util/assistant · Indexed

Assistant Discussion component in chat, planning & games.

Methods: json(), clean(), lower(), strings()

AssistantEditDialogue.kt

util/assistant · Indexed

Assistant Edit Dialogue component in chat, planning & games.

Methods: request(), field(), priority(), toggle()

AssistantFeatureActions.kt

util/assistant · Indexed

Assistant Feature Actions component in chat, planning & games.

Methods: encode(), decode(), encode(), decode()

AssistantFeatureLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Assistant Feature.

Methods: clock(), window(), days(), timeText()

AssistantFeatures.kt

util/assistant · Indexed

Assistant Features component in chat, planning & games.

Methods: appList(), answer(), chest()

AssistantFocusLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Assistant Focus.

Methods: request()

AssistantGuidedPlanning.kt

util/assistant · Indexed

Assistant Guided Planning component in chat, planning & games.

Methods: get(), domain(), startRequest(), isRequest()

AssistantHabitConversation.kt

util/assistant · Indexed

Assistant Habit Conversation component in chat, planning & games.

Methods: proposedValue(), schedule(), information(), revise()

AssistantHabitEditing.kt

util/assistant · Indexed

Assistant Habit Editing component in chat, planning & games.

Methods: request(), match(), value(), hint()

AssistantHabitLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Assistant Habit.

Methods: report()

AssistantInitiative.kt

util/assistant · Indexed

Assistant Initiative component in chat, planning & games.

Methods: prefs(), dismiss(), attach()

AssistantInterpretation.kt

util/assistant · Indexed

Assistant Interpretation component in chat, planning & games.

Methods: interpret(), explanation()

AssistantLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Assistant.

Methods: unwrap(), actionGuard(), acknowledgement()

AssistantLearningFeatures.kt

util/assistant · Indexed

Assistant Learning Features component in chat, planning & games.

Methods: validSummary(), answer(), score()

AssistantLearningLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Assistant Learning.

Methods: command(), toJson(), fromJson(), command()

AssistantPersonalization.kt

util/assistant · Indexed

Assistant Personalization component in chat, planning & games.

Methods: prefs(), tone(), quiet(), suggestions()

AssistantPlanStore.kt

util/assistant · Reviewed

Assistant plan persistence/commit boundary with existing task integration.

Methods: undo(), hash(), save(), effects()

AssistantPlanning.kt

util/assistant · Indexed

Assistant Planning component in chat, planning & games.

Methods: live(), toJson(), fromJson(), clean()

AssistantPlanningArchive.kt

util/assistant · Indexed

Assistant Planning Archive component in chat, planning & games.

Methods: withoutAuthority(), export(), preflight(), restore()

AssistantPreferences.kt

util/assistant · Indexed

Assistant Preferences component in chat, planning & games.

Methods: find(), read(), validate(), extractValue()

AssistantPresentation.kt

util/assistant · Indexed

Assistant Presentation component in chat, planning & games.

Methods: toJson(), fromJson(), allows(), data()

AssistantReplyStyle.kt

util/assistant · Indexed

Assistant Reply Style component in chat, planning & games.

Methods: prefs(), concise(), emoji(), preference()

AssistantResponseContent.kt

util/assistant · Indexed

Assistant Response Content component in chat, planning & games.

Methods: attach(), setting(), taskList(), tasks()

AssistantRoutineConversation.kt

util/assistant · Indexed

Assistant Routine Conversation component in chat, planning & games.

Methods: remember(), reference(), summary(), change()

AssistantRoutineDialogue.kt

util/assistant · Indexed

Assistant Routine Dialogue component in chat, planning & games.

Methods: hasDraft(), answer(), preview(), hint()

AssistantSession.kt

util/assistant · Reviewed

Conversation references, pending actions, confirmations and protected session persistence.

Methods: focusTask(), focusTasks(), rememberTaskList(), clearSuspended()

AssistantSocial.kt

util/assistant · Indexed

Assistant Social component in chat, planning & games.

Methods: frame(), features(), hasInterestOpening(), interestOpening()

AssistantStateLock.kt

util/assistant · Reviewed

Serializes assistant state access; complements database transactions and maintenance gates.

Methods:

AssistantTaskDateRepair.kt

util/assistant · Indexed

Assistant Task Date Repair component in chat, planning & games.

Methods: clockOnly(), repair(), unclear(), preserveTime()

AssistantTaskLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Assistant Task.

Methods: normalize(), request(), match(), select()

AssistantTaskReferences.kt

util/assistant · Indexed

Assistant Task References component in chat, planning & games.

Methods: select(), unclear(), resolve(), position()

AssistantTaskViews.kt

util/assistant · Indexed

Assistant Task Views component in chat, planning & games.

Methods: answer()

AssistantTranscriptFeatures.kt

util/assistant · Indexed

Assistant Transcript Features component in chat, planning & games.

Methods: drafts(), answer()

AssistantVocabularyFeatures.kt

util/assistant · Indexed

Assistant Vocabulary Features component in chat, planning & games.

Methods: features(), get(), recordCard(), answer()

AyvaPersonality.kt

util/assistant · Indexed

Ayva Personality component in chat, planning & games.

Methods: tone(), preference(), answer()

CanonicalPlanTasks.kt

util/assistant · Indexed

Canonical Plan Tasks component in chat, planning & games.

Methods: canonicalWorkSessions(), occupiedWork()

ChatExperiences.kt

util/assistant · Reviewed

Focused offline experiences: games, facts, jokes, encouragement, stories and resumable experience state.

Methods: unsupportedTopic(), help()

ChatMode.kt

util/assistant · Indexed

Chat Mode component in chat, planning & games.

Methods: suggested(), activate(), get(), set()

ConversationKnowledge.kt

util/assistant · Indexed

Conversation Knowledge component in chat, planning & games.

Methods: match(), find(), explain()

ConversationLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Conversation.

Methods: space(), normalize(), unquote(), list()

ConversationMeaning.kt

util/assistant · Indexed

Conversation Meaning component in chat, planning & games.

Methods: clauses(), kind()

ConversationPlanStore.kt

util/assistant · Reviewed

Persists/commits planning workspaces with canonical task links and operation receipts.

Methods: clearAll(), all(), active(), document()

ConversationPlanner.kt

util/assistant · Indexed

Conversation Planner component in chat, planning & games.

Methods: respond(), signature(), missingDiscussionGoal(), edit()

ConversationReasoning.kt

util/assistant · Indexed

Conversation Reasoning component in chat, planning & games.

Methods: json(), key(), opposite(), read()

ConversationRegistry.kt

util/assistant · Indexed

Conversation Registry component in chat, planning & games.

Methods: answer()

ConversationRouting.kt

util/assistant · Indexed

Conversation Routing component in chat, planning & games.

Methods: nativeRequest(), candidates(), winner(), permits()

ConversationStories.kt

util/assistant · Indexed

Conversation Stories component in chat, planning & games.

Methods: answer()

ConversationVoice.kt

util/assistant · Indexed

Conversation Voice component in chat, planning & games.

Methods: questions(), concise(), preference(), finish()

DiscussionComparison.kt

util/assistant · Indexed

Discussion Comparison component in chat, planning & games.

Methods: display(), json(), same(), json()

DiscussionCriteria.kt

util/assistant · Indexed

Discussion Criteria component in chat, planning & games.

Methods: hasData(), json(), retain(), value()

EntertainmentPack.kt

util/assistant · Indexed

Entertainment Pack component in chat, planning & games.

Methods:

EverydayTopics.kt

util/assistant · Indexed

Everyday Topics component in chat, planning & games.

Methods: answer()

ExistingTaskPlanning.kt

util/assistant · Indexed

Existing Task Planning component in chat, planning & games.

Methods: adopt()

ExperienceContent.kt

util/assistant · Reviewed

Authored finite entertainment content, source-linked facts and original traditional-fable retellings.

Methods:

FocusedChat.kt

util/assistant · Indexed

Focused Chat component in chat, planning & games.

Methods: values(), response(), rememberActivity(), answer()

GeneralConversation.kt

util/assistant · Indexed

General Conversation component in chat, planning & games.

Methods: root(), current(), strings(), archive()

GeneralDialogue.kt

util/assistant · Reviewed

Coordinates bounded general-dialogue interpretation and fallback; not unrestricted model-based conversation.

Methods: json(), json(), values(), has()

GeneralDialoguePacks.kt

util/assistant · Indexed

General Dialogue Packs component in chat, planning & games.

Methods: elaborations(), explanation(), suggestions(), reflect()

MilestonePlanning.kt

util/assistant · Indexed

Milestone Planning component in chat, planning & games.

Methods: matches(), scope(), overview(), answer()

NativeGeneralDialogue.kt

util/assistant · Reviewed

Optional pinned ChatScript subprocess adapter; trusted packaged rules only, same app UID, Kotlin fallback.

Methods: eligible(), decode(), interpret(), readBytesBounded()

OfflineAssistant.kt

util/assistant · Reviewed

Main offline assistant turn coordinator; selects supported modes/features and guarded action proposals.

Methods: token(), respond(), invalidatePlanReview(), rejectInput()

PlanBreakdown.kt

util/assistant · Indexed

Plan Breakdown component in chat, planning & games.

Methods: method(), answer(), eligible(), validate()

PlanChangeTrail.kt

util/assistant · Indexed

Plan Change Trail component in chat, planning & games.

Methods: record(), describe()

PlanClock.kt

util/assistant · Indexed

Plan Clock component in chat, planning & games.

Methods: format(), boundary()

PlanCollaboration.kt

util/assistant · Indexed

Plan Collaboration component in chat, planning & games.

Methods: answer()

PlanConversationRepairs.kt

util/assistant · Indexed

Plan Conversation Repairs component in chat, planning & games.

Methods: parse()

PlanDiff.kt

util/assistant · Indexed

Plan Diff component in chat, planning & games.

Methods: describe()

PlanLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Plan.

Methods: normalize(), command(), startsPlanning(), domain()

PlanMilestone.kt

util/assistant · Indexed

Plan Milestone component in chat, planning & games.

Methods: json(), validate(), readList(), reconcile()

PlanPolicy.kt

util/assistant · Indexed

Plan Policy component in chat, planning & games.

Methods: validate(), budget(), json(), allows()

PlanProgress.kt

util/assistant · Indexed

Plan Progress component in chat, planning & games.

Methods: answer()

PlanTemporalLanguage.kt

util/assistant · Indexed

Recognizes supported language patterns for Plan Temporal.

Methods: expression()

PlanWorkAlternatives.kt

util/assistant · Indexed

Plan Work Alternatives component in chat, planning & games.

Methods: laterOnSameDay(), withinAvailability()

PlanWorkspace.kt

util/assistant · Indexed

Plan Workspace component in chat, planning & games.

Methods: json(), read(), withSessions(), shifted()

PlanningDialogue.kt

util/assistant · Indexed

Planning Dialogue component in chat, planning & games.

Methods: json(), read(), strings(), goal()

PlanningDraft.kt

util/assistant · Indexed

Planning Draft component in chat, planning & games.

Methods: json(), json(), tasks(), read()

PlanningLibrary.kt

util/assistant · Indexed

Planning Library component in chat, planning & games.

Methods: empty(), read(), write(), clear()

PlanningMethods.kt

util/assistant · Indexed

Planning Methods component in chat, planning & games.

Methods: groups(), method(), title(), domain()

SavedPlanHistory.kt

util/assistant · Indexed

Saved Plan History component in chat, planning & games.

Methods: clear(), read(), write(), document()

SharedPlanConstraints.kt

util/assistant · Indexed

Shared Plan Constraints component in chat, planning & games.

Methods: validate(), group(), answer(), apply()

TaskScheduleView.kt

util/assistant · Indexed

Task Schedule View component in chat, planning & games.

Methods: matches(), describe()

TripChecklistConversation.kt

util/assistant · Reviewed

Trip/exam checklist conversation state, incremental additions/edits, dates, summaries and completion handling.

Methods: json(), json(), validate(), read()

TripPlanningBridge.kt

util/assistant · Indexed

Trip Planning Bridge component in chat, planning & games.

Methods: advancedRequest(), migrate(), promote(), reconcile()

AyvaChatCommands.kt

util/command · Indexed

Ayva Chat Commands component in chat, planning & games.

Methods: talkQuery()

AyvaCompoundCommandHandler.kt

util/command · Indexed

Ayva Compound Command Handler component in chat, planning & games.

Methods: splitCompoundCommands(), queryAppUsageTelemetry(), extractTimeLimitMinutes()

AyvaContextEngine.kt

util/context · Indexed

Core transformation or evaluation routines for Ayva Context.

Methods: determineCircadianPhase(), captureSnapshot()

AyvaIntentRouter.kt

util/router · Reviewed

Routes interpreted input toward task/native actions/Talk paths; review precedence alongside OfflineAssistant.

Methods: route(), containsPhrase(), handleSlashCommand(), findDuplicateTask()

Voice capture & transcript editing · 7 files

VoiceRecordDialog.kt

ui/screens/notes · Indexed

Dialog interface for Voice Record.

Methods: VoiceRecordDialog()

VoiceCaptureContent.kt

ui/speech · Reviewed

Shared transcription popup UI, editable draft transcript and capture controls.

Methods: VoiceCaptureContent(), TranscriptText()

VoiceStyle.kt

ui/speech · Indexed

Voice Style component in voice capture & transcription.

Methods: fromId(), read(), save()

VoiceStyleSettings.kt

ui/speech · Indexed

Voice Style Settings component in voice capture & transcription.

Methods: VoiceStyleSettingsRow(), VoiceStylePicker(), VoiceStylePreview()

VoiceVisualization.kt

ui/speech · Reviewed

Renders voice amplitude/motion visuals; this is presentation rather than speech recognition.

Methods: rememberVoiceMotionAllowed(), readMotion(), onChange(), VoiceVisualization()

OnDeviceTranscriber.kt

util/speech · Reviewed

Uses the default phone SpeechRecognizer; despite the name, the selected provider may use online recognition.

Methods: finalizedTranscript(), startRecording(), retryRecording(), beginRecording()

SpeechServiceDiagnostics.kt

util/speech · Reviewed

Reports which speech services are available/configured; diagnostic Google package visibility is not bundled Google code.

Methods: describe(), read()

Arithmetic, drills & vocabulary · 34 files

DictionaryBookmark.kt

data · Indexed

Dictionary Bookmark component in learning & vocabulary.

Methods: get(), insert()

VocabDao.kt

data · Indexed

Database query interface for Vocab.

Methods: findIdioms(), findOws(), insertDictionaryOws(), getBookmarkedIdioms()

VocabDatabase.kt

data · Reviewed

Room vocabulary database seeded from vocab.db; distinct from the large Wordset shard dictionary.

Methods: migrate(), migrate()

VocabEntities.kt

data · Indexed

Vocab Entities component in learning & vocabulary.

Methods:

VocabRepository.kt

data · Indexed

Data access and update boundary for Vocab.

Methods: getNextIdiomToLearn(), getNextOwsToLearn(), markIdiomLearned(), markOwsLearned()

DrillDatabase.kt

data/drill · Reviewed

App-private Room SQLite drill-session summaries; separate from the encrypted notes/focus stores.

Methods: insertSession(), getSessionById(), getAllSessions(), getAllSessionsSync()

DrillModels.kt

data/drill · Indexed

Drill Models component in learning & vocabulary.

Methods: toJson(), fromJson(), toJson(), fromJson()

AptitudeReminderHelper.kt

service · Indexed

Supporting functions for Aptitude Reminder.

Methods: parseHourAndMinute()

AptitudeReminderReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Aptitude Reminder.

Methods: scheduleDrillReminders(), scheduleRandomDrillReminders(), cancelAllReminders(), cancelExactDrillReminder()

VocabBriefContent.kt

ui/components · Indexed

Vocab Brief Content component in learning & vocabulary.

Methods: VocabBriefContent(), VocabItemCard()

VocabRetentionHubChatCard.kt

ui/components · Indexed

Vocab Retention Hub Chat Card component in learning & vocabulary.

Methods: computeMasteryProgress(), VocabRetentionHubChatCard(), LearnedWordCardItem()

AptitudeCardHelper.kt

ui/screens · Indexed

Supporting functions for Aptitude Card.

Methods: clampAccuracy(), clampStreakFreezes()

AptitudeProfileCard.kt

ui/screens · Indexed

Aptitude Profile Card component in learning & vocabulary.

Methods: AptitudeProfileCard(), CompactStatItem(), CompactDivider()

ArithmeticCard.kt

ui/screens · Indexed

Arithmetic Card component in learning & vocabulary.

Methods: parseDrillJson(), computeProgressFraction(), ArithmeticCard()

DrillSummaryCard.kt

ui/screens · Indexed

Drill Summary Card component in learning & vocabulary.

Methods: DrillSummaryCard()

FullscreenDrillSummaryView.kt

ui/screens · Indexed

Fullscreen Drill Summary View component in learning & vocabulary.

Methods: FullscreenDrillSummaryView()

FullscreenDrillView.kt

ui/screens · Indexed

Fullscreen Drill View component in learning & vocabulary.

Methods: FullscreenDrillView()

FullscreenSolutionsView.kt

ui/screens · Indexed

Fullscreen Solutions View component in learning & vocabulary.

Methods: FullscreenSolutionsView()

DrillBottomActionRow.kt

ui/screens/drill · Indexed

Drill Bottom Action Row component in learning & vocabulary.

Methods: DrillBottomActionRow()

DrillPaletteHelper.kt

ui/screens/drill · Indexed

Supporting functions for Drill Palette.

Methods: getSafeCount()

DrillQuestionPaletteModal.kt

ui/screens/drill · Indexed

Drill Question Palette Modal component in learning & vocabulary.

Methods: DrillQuestionPaletteModal(), PaletteLegendItem()

DrillStreakHelper.kt

ui/screens/drill · Indexed

Supporting functions for Drill Streak.

Methods: calculateCycleDay()

DrillSummaryMathHelper.kt

ui/screens/drill · Indexed

Supporting functions for Drill Summary Math.

Methods: calculateAccuracy()

DrillTopBar.kt

ui/screens/drill · Indexed

Drill Top Bar component in learning & vocabulary.

Methods: DrillTopBar(), DrillComboStreakCapsule()

DrillTopBarHelper.kt

ui/screens/drill · Indexed

Supporting functions for Drill Top Bar.

Methods: formatBlitzTimer(), calculateBlitzProgress()

ExplanationSection.kt

ui/screens/drill · Indexed

Explanation Section component in learning & vocabulary.

Methods: ExplanationSection()

FullscreenDrillHelper.kt

ui/screens/drill · Indexed

Supporting functions for Fullscreen Drill.

Methods: updateCombo(), safeCorrectIndex()

SolutionCardItem.kt

ui/screens/drill · Indexed

Solution Card Item component in learning & vocabulary.

Methods: SolutionCardItem()

SolutionsTopBar.kt

ui/screens/drill · Indexed

Solutions Top Bar component in learning & vocabulary.

Methods: SolutionsTopBar(), SolutionsQuestionNumberStrip(), FilterTabPill(), formatSecondsToMinutesSec()

SolutionsViewHelper.kt

ui/screens/drill · Indexed

Supporting functions for Solutions View.

Methods: parseQuestionsSafely(), safeQuestionIndex()

AptitudeManager.kt

util · Reviewed

Learning profile, drill statistics and progress persistence.

Methods: init(), isVacationMode(), setVacationMode(), getDailyDrillKey()

ArithmeticEngine.kt

util · Reviewed

Generates/evaluates supported arithmetic learning questions and explanations.

Methods: generateQuestion(), generateSpeedMath(), generateSimplification(), generateApproximation()

DrillSummaryHelper.kt

util · Indexed

Supporting functions for Drill Summary.

Methods: generateSummaryMessage()

VocabQuizChoices.kt

util · Indexed

Vocab Quiz Choices component in learning & vocabulary.

Methods: key(), build()

Profile, economy, quests & Pro · 11 files

DuolingoMysteryChestDialog.kt

ui/components · Reviewed

Native chest dialog, reveal lifecycle and reward presentation.

Methods: nextTier(), rollRarity(), DuolingoChest3DGraphic(), DuolingoMysteryChestDialog()

MysteryBoxChatCard.kt

ui/components · Indexed

Mystery Box Chat Card component in profile, rewards & pro.

Methods: MysteryBoxChatCard()

SmokePawChestScene.kt

ui/components · Reviewed

Private offline HTML/vector rasterizer bridge for the approved tiered chest animation.

Methods: durationMs(), decode(), SmokePawChestScene(), shouldOverrideUrlLoading()

AccountScreen.kt

ui/screens · Indexed

User-interface screen for Account.

Methods: rememberEnclaveColors(), AccountScreen(), EnclaveTopBar(), EnclaveIdentityHeroCard()

DailyQuestsCard.kt

ui/screens · Indexed

Daily Quests Card component in profile, rewards & pro.

Methods: DailyQuestsCard()

SubscriptionScreen.kt

ui/screens · Indexed

User-interface screen for Subscription.

Methods: SubscriptionScreen(), ProStatusCard(), ActivationSection(), ProBenefitsSection()

DailyQuestManager.kt

util · Reviewed

Quest/chest eligibility and award calculation; visual Skip/Collect must not roll another reward.

Methods: init(), getTodayDateString(), refreshState(), recordDrillCompleted()

FocusEconomyManager.kt

util · Reviewed

Gold/level/avatar economy and profile reward state.

Methods: xpCurrentLevelBase(), xpNextLevelThreshold(), xpGainedInCurrentLevel(), xpNeededForNextLevel()

GamificationHaptics.kt

util · Indexed

Gamification Haptics component in profile, rewards & pro.

Methods: getVibrator(), playCorrect(), playWrong(), playCombo()

LicenseManager.kt

util · Reviewed

Local signed lifetime-Pro token verification; no Google Play Billing SDK is used by this component.

Methods: init(), verifyAndActivateLicense(), activatePro(), decodeBase64Url()

ProfileAvatarManager.kt

util · Indexed

Coordinates operations and state for Profile Avatar.

Methods: getAvatar(), getAvatarImageRes(), getAvatarBorderColor(), getAvatarTitle()

Settings, themes & Vizag scenery · 30 files

AyvaNotificationColorsDialog.kt

ui/components · Indexed

Dialog interface for Ayva Notification Colors.

Methods: AyvaNotificationColorsDialog(), CategoryCompactRow(), getCategoryIcon(), getCategorySubtitle()

DevicePermissionsState.kt

ui/components · Indexed

Device Permissions State component in settings, themes & permissions.

Methods: rememberDevicePermissions()

LayeredVizagDrawable.kt

ui/components · Reviewed

Native independent-motion renderer for the five replaced Vizag scenes.

Methods: objects(), route(), run(), start()

LegacyWebpDrawable.kt

ui/components · Reviewed

Playback path for retained older animated WebP scenes.

Methods: u24(), u32(), tag(), render()

PermissionsDialog.kt

ui/components · Indexed

Dialog interface for Permissions.

Methods: SetupPermissionsDialog(), PermissionSetupCard()

SettingsMenuRow.kt

ui/components · Indexed

Settings Menu Row component in settings, themes & permissions.

Methods: SettingsMenuRow(), SettingsMenuSectionHeader(), SettingsMenuPage(), SettingsMenuLink()

SettingsPreferences.kt

ui/components · Indexed

Settings Preferences component in settings, themes & permissions.

Methods: writeSetting(), rememberSetting(), read()

SettingsSearch.kt

ui/components · Indexed

Settings Search component in settings, themes & permissions.

Methods: matches(), SettingsSearchDestination(), settingsSearchTarget(), SettingsSearchSheet()

SettingsSearchCatalog.kt

ui/components · Reviewed

Canonical settings labels/aliases and destinations for in-app search.

Methods: resolve(), id(), routeFor()

SettingsVizagFooter.kt

ui/components · Reviewed

Chooses/hosts the nine Vizag scenery variants and signature link.

Methods: decodeVizagArtwork(), claimNext(), rememberVizagScene(), SettingsVizagFooter()

BubbleSettingsHelper.kt

ui/screens · Indexed

Supporting functions for Bubble Settings.

Methods: coerceSliderValue(), parseTimeSafe()

BubbleSettingsScreen.kt

ui/screens · Indexed

User-interface screen for Bubble Settings.

Methods: BubbleSettingsScreen(), notifyService(), LiveBubblePreviewCard(), SettingsSliderRow()

HelpSettingsScreen.kt

ui/screens · Indexed

User-interface screen for Help Settings.

Methods: HelpSettingsScreen(), setup(), DiagnosticsSettingsScreen()

PreferencesHubHelper.kt

ui/screens · Indexed

Supporting functions for Preferences Hub.

Methods: getTaskDao()

PreferencesHubScreen.kt

ui/screens · Indexed

User-interface screen for Preferences Hub.

Methods: PreferencesHubScreen()

SecurityScreen.kt

ui/screens · Indexed

User-interface screen for Security.

Methods: SecurityScreen(), GrantedBadge(), GrantButton(), SafetySnapshotsBottomSheet()

SettingsScreen.kt

ui/screens · Indexed

User-interface screen for Settings.

Methods: SettingsScreen(), SettingsSectionHeader(), SettingsNavigationRow(), SettingsStepperRow()

SettingsScreenHelper.kt

ui/screens · Indexed

Supporting functions for Settings Screen.

Methods: createWidgetConfigureIntent(), getNextReminderInterval(), coerceSoftLockDuration(), coerceSoftUnlockDuration()

WidgetsSettingsScreen.kt

ui/screens · Indexed

User-interface screen for Widgets Settings.

Methods: WidgetsSettingsScreen(), configure()

BackupDialogs.kt

ui/screens/security · Indexed

Backup Dialogs component in settings, themes & permissions.

Methods: ExportBackupPasswordDialog(), RestoreBackupPasswordDialog(), ConfigureRestoredVaultPinDialog()

ScreenCaptureProtection.kt

ui/security · Indexed

Screen Capture Protection component in settings, themes & permissions.

Methods: installScreenCaptureProtection(), refresh(), onResume(), onDestroy()

Color.kt

ui/theme · Indexed

Color component in settings, themes & permissions.

Methods:

Theme.kt

ui/theme · Indexed

Theme component in settings, themes & permissions.

Methods: findActivity(), FocusByRjTheme()

Type.kt

ui/theme · Indexed

Type component in settings, themes & permissions.

Methods:

AppIconManager.kt

util · Indexed

Coordinates operations and state for App Icon.

Methods: init(), getCurrentIcon(), setAppIcon()

AppThemeManager.kt

util · Indexed

Coordinates operations and state for App Theme.

Methods: fromId(), fromId(), init(), setTheme()

AyvaAlertTheme.kt

util · Indexed

Ayva Alert Theme component in settings, themes & permissions.

Methods: getPresetById(), getPresetByBgHex(), isEnabled(), setEnabled()

DevicePermissions.kt

util · Indexed

Device Permissions component in settings, themes & permissions.

Methods: granted(), request(), read()

PermissionUtils.kt

util · Indexed

Permission Utils component in settings, themes & permissions.

Methods: hasUsageStatsPermission(), requestUsageStatsPermission(), hasOverlayPermission(), requestOverlayPermission()

SettingsPreferenceRules.kt

util · Indexed

Settings Preference Rules component in settings, themes & permissions.

Methods: nextReminderInterval(), datePreview(), routeFor(), taskStyleValue()

Launcher widgets & compact editors · 19 files

ExamCountdown.kt

widget · Reviewed

Countdown model/validation and portable event storage; launcher widget IDs remain local bindings.

Methods: target(), validate(), headline(), progress()

ExamCountdownConfigureActivity.kt

widget · Indexed

Android window / activity entry point for Exam Countdown Configure.

Methods: onCreate(), ExamCountdownConfigScreen(), change(), pickDate()

ExamCountdownProvider.kt

widget · Reviewed

Countdown widget rendering, clock/date updates and pin callback handling.

Methods: onUpdate(), onAppWidgetOptionsChanged(), onDeleted(), onDisabled()

NoteWidgetActionReceiver.kt

widget · Indexed

Handles Android broadcasts or scheduled callbacks for Note Widget Action.

Methods: onReceive()

NoteWidgetConfig.kt

widget · Indexed

Configuration models and persistence for Note Widget.

Methods: next(), fromNameOrDefault(), getConfig(), getStringVal()

NoteWidgetConfigureActivity.kt

widget · Indexed

Android window / activity entry point for Note Widget Configure.

Methods: onCreate(), NoteWidgetConfigScreen(), NoteWidgetLivePreview()

NoteWidgetDrawableGenerator.kt

widget · Indexed

Note Widget Drawable Generator component in home-screen widgets.

Methods: getWidgetColors(), createWidgetBackground(), removeEldestEntry(), createCheckboxBitmap()

NoteWidgetProvider.kt

widget · Reviewed

Updates the note widget RemoteViews and attaches actions/content adapters.

Methods: updateAllWidgets(), updateWidget(), updateWidgetInternal(), onUpdate()

NoteWidgetService.kt

widget · Indexed

Note Widget Service component in home-screen widgets.

Methods: onGetViewFactory(), onCreate(), onDataSetChanged(), loadData()

NoteWidgetTypography.kt

widget · Reviewed

Explicit note-widget text sizing rules; avoids auto-sizing surprises in RemoteViews.

Methods: apply(), applyHeader(), adapterIntent()

QuickAddNoteItemActivity.kt

widget · Indexed

Android window / activity entry point for Quick Add Note Item.

Methods: onCreate(), QuickAddNoteItemDialog(), startVoiceDictation(), performSubmit()

QuickEditNoteActivity.kt

widget · Indexed

Android window / activity entry point for Quick Edit Note.

Methods: onCreate(), QuickEditNoteOverlay(), startVoiceDictation(), saveNote()

TodoWidgetActionReceiver.kt

widget · Indexed

Handles Android broadcasts or scheduled callbacks for Todo Widget Action.

Methods: onReceive()

TodoWidgetConfigureActivity.kt

widget · Indexed

Android window / activity entry point for Todo Widget Configure.

Methods: onCreate(), WidgetConfigScreen(), WidgetLivePreview(), PreviewTaskRow()

TodoWidgetProvider.kt

widget · Reviewed

Updates task widget RemoteViews, tabs and row actions.

Methods: getSelectedTab(), setSelectedTab(), updateAllWidgets(), updateWidget()

TodoWidgetService.kt

widget · Indexed

Todo Widget Service component in home-screen widgets.

Methods: onGetViewFactory(), onCreate(), onDataSetChanged(), loadTasks()

WidgetConfig.kt

widget · Indexed

Configuration models and persistence for Widget.

Methods: fromNameOrDefault(), getConfig(), getStr(), getInt()

WidgetDrawableGenerator.kt

widget · Indexed

Widget Drawable Generator component in home-screen widgets.

Methods: removeEldestEntry(), removeEldestEntry(), createWidgetBackground(), createItemBackground()

WidgetEditorComponents.kt

widget · Reviewed

Shared compact customization rows, radio-choice dialogs, sliders, tabs and optional live previews.

Methods: WidgetEditor(), WidgetEditorRow(), WidgetChoiceRow(), WidgetToggleRow()

Android lifetime, notifications & diagnostics · 13 files

BootReceiver.kt

service · Reviewed

Restores optional services and enqueues reminder reconciliation after boot or app update.

Methods: onReceive(), attempt()

BubbleService.kt

service · Reviewed

Floating chat bubble: foreground lifetime, snooze/visibility, fullscreen/landscape hiding and recovery.

Methods: onDisplayAdded(), onDisplayRemoved(), onDisplayChanged(), onReceive()

DailySummaryReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Daily Summary.

Methods: scheduleDailySummaries(), scheduleSingleSummary(), computeNextTriggerTime(), onReceive()

PackageChangeReceiver.kt

service · Indexed

Handles Android broadcasts or scheduled callbacks for Package Change.

Methods: onReceive()

ReminderClockReceiver.kt

service · Reviewed

Responds to local time/timezone changes by scheduling reminder reconciliation.

Methods: onReceive()

ReminderOverlayStore.kt

service · Reviewed

Persists/reconstructs pending reminder overlay presentation state.

Methods: read(), write()

ReminderReconcileWorker.kt

service · Reviewed

Rebuilds reminder schedules, isolating failures; calendar changes preserve absolute task timestamps.

Methods: doWork(), attempt(), enqueue()

UnifiedOverlayCoordinator.kt

service · Reviewed

Queues competing reminder overlays so task/habit windows do not replace one another unpredictably.

Methods: record(), savePending(), initialize(), restorePending()

SafePrefsExtensions.kt

util · Indexed

Safe Prefs Extensions component in android services & notifications.

Methods: getSafeLong(), getSafeInt(), getSafeFloat(), getSafeBoolean()

TalkAction.kt

util · Indexed

Talk Action component in android services & notifications.

Methods: executeAsync(), execute()

AppLogger.kt

util/diagnostics · Indexed

App Logger component in android services & notifications.

Methods: init(), setRecordingEnabled(), isRecording(), v()

CrashHandler.kt

util/diagnostics · Indexed

Crash Handler component in android services & notifications.

Methods: uncaughtException(), recordCrashToDisk(), init()

DiagnosticManager.kt

util/diagnostics · Indexed

Coordinates operations and state for Diagnostic.

Methods: isRecordingEnabled(), setRecordingEnabled(), getLogsDirectory(), getExportsDirectory()

Sources, scope & handover

This atlas describes commit cce930915554c64c4884b134c238daed93e78155 and version 2.0.37. It inventories tracked source, resources, assets, tests and tools. Build outputs, local signing credentials, user data, caches and existing untracked mockups are excluded. No private keys, secrets or real user databases are copied into the guide.

What this guide does and does not establish

- The inventory is complete for git-tracked files at this snapshot. Reviewed diagrams explain architecture; long-tail file roles and static coupling are extracted/inferred and labelled. Reflection, dynamic dispatch, generated code, resource lookup and runtime-only relationships can be missing from static edges.
- A matching upstream notice and pinned source provide stronger origin evidence than a filename or prior research plan. Unknown material is explicitly left unknown; direct copying of Notesnook/Duolingo/Bare Browser is not established here.
- License family summaries are preparation guidance. This is not a legal opinion, an exhaustive source-similarity audit, a complete release-APK SBOM or an official F-Droid acceptance decision.
- Primary policy pages were reviewed for this assessment; terms and scanners can change. Item-specific historical download terms may differ from current website terms.
- Historical screenshots/docs are labelled and may differ from device state. Generating this atlas does not redesign app screens or change app behavior.

Primary external sources

1. F-Droid inclusion policy — Policy for free software, redistributable assets, source/build expectations and maturity.
   https://f-droid.org/en/docs/Inclusion_Policy/
2. F-Droid anti-features — NonFreeAssets and other declarations; not a blanket exemption for non-free code.
   https://f-droid.org/en/docs/Anti-Features/
3. F-Droid reproducible builds — Reproducible build/signature-copy route; no private-key sharing.
   https://f-droid.org/en/docs/Reproducible_Builds/
4. F-Droid submission guide — Submission and metadata workflow.
   https://f-droid.org/en/docs/Submitting_to_F-Droid_Quick_Start_Guide/
5. F-Droid scanner source — Scanner rules; inspected policy/source, not an official scan of this app.
   https://gitlab.com/fdroid/fdroidserver/-/blob/master/fdroidserver/scanner.py
6. Wordset Dictionary — Confirmed pinned data source; CC BY-SA/WordNet notices.
   https://github.com/wordset/wordset-dictionary/tree/4503bfafbe4556361a296b9d368c18168dfb56c3
7. ChatScript upstream — Confirmed pinned native source.
   https://github.com/ChatScript/ChatScript/tree/9f5eec4736ba22bd992a6498c1e0052e2a795125
8. Rive terms — Current Community Content terms C.6; exact item provenance still needed.
   https://rive.app/docs/legal/terms-of-service
9. Lottie Simple License — Current item-license framework; confirm the actual downloaded file and terms at acquisition.
   https://lottiefiles.com/page/license
10. Rive Android runtime — Runtime code license is separate from downloaded .riv art.
    https://github.com/rive-app/rive-android
11. Lottie Android runtime — Apache-licensed player; does not license third-party animation content.
    https://github.com/airbnb/lottie-android
12. Creative Commons BY-SA 4.0 — Dictionary data obligations.
    https://creativecommons.org/licenses/by-sa/4.0/

13. Creative Commons BY 4.0 — Attribution/source/license/change requirements for qualifying Rive community art.
    https://creativecommons.org/licenses/by/4.0/
14. SIL Open Font License — Font-specific license family; match embedded actual-file terms.
    https://openfontlicense.org/

Key source evidence inside this repository

LICENSE

TRADEMARK.md

build.gradle.kts

AndroidManifest.xml

DICTIONARY_DATA.md

README.md

Files in the handover

FILE	USE
`RUN-Visual-Guide.html`	Self-contained interactive atlas. Open locally; no server, install or internet needed to browse.
`RUN-Visual-Guide.pdf`	Printable reading copy with diagrams and the full production-code catalog.
`RUN-file-index.csv`	Every tracked file, role, origin evidence, license status, pinned source URL, size and SHA-256.
`RUN-code-connections.csv`	All extracted static source edges and their evidence type.
`RUN-material-ledger.csv`	Animation/font/seed rights checklist with fields for recovering original sources.
`atlas-data.json`	Machine-readable inventory with declarations, methods, edges and manifest data.
`atlas-validation.json`	Checks of documentation coverage and rendered artifacts.
`README.txt`	Quick start, scope and rebuild instructions.
`source/ and diagrams/`	Generator, reviewed narrative and editable Graphviz maps for future updates.